The level–60 computation for Erdős Problem #307: a certificate, a split sieve, and the pair sector

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Abstract

A solution of Erdős Problem #307 is a two–cycle of the arithmetic derivative, and any solution has $|P \cup Q| \geq 60$. This note records the computation that decides the level–60 families. The certificate of Proposition 1.2 empties 31,219 of the 49,961 admissible bases by a Jacobi symbol, and trial division to 10^6 leaves 7,713. For the rest we give the split sieve: writing a cycle on a family $S \cup M$ as $a = \prod(T \cup M)$, $b = \prod(S \setminus T)$, the two cycle equations give $\prod T \mid \partial(\prod(S \setminus T))$ together with an identity free of the tail. The divisibility is the identity read modulo $\prod T$, so it is a consequence rather than a hypothesis, and by the anatomy lemma it is a congruence at each prime of T, costing a factor $1/r$ each; $\prod_{p \in S}(1 + 1/p)$ subsets survive out of $2^{|S|}$. Neither statement mentions the cofactor A_S, so the criterion reaches the families whose A_S is composite and, the tail being arbitrary, the pair sector, both of which the arity–one method cannot. Every level–60 base surviving the certificate is decided over all T with $|T| \leq 6$: about 9×10^{11} splits, no factorisation, and no two–cycle. The heuristic model and a breeder form for certificate search are included. The range $|T| \geq 7$ is not decided and its price is stated.

1 The certificate programme and its anatomy

The barrier of Section 4 of the core note bounds a solution; it does not exclude one. This section records the programme that attempted the exclusion level by level, its completeness, and the point at which it provably stops. Nothing here is used by the results of the core note.

Remark 1.1 (Verification tier, and why the ladder stops here)**.** Theorem 4.3 is machine–checked in Lean 4, and so is Proposition 4.5: the theorem `erdos307_sixty` formalises the entire argument; the forced–primes and element bounds, the Pythagorean plus–square, and a pruned depth–first search over the pool whose *completeness is proved, not assumed* (`dfs_sound`), the search itself running under the kernel; on top of the two independent unverified programs. The level cannot be climbed further by finite means: already at $|U| = 60$ the family $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$ qualifies for *every* prime $q > 277$, and on it the two square conditions read $x^2 = Aq + \Pi_{59}$, $y^2 = Bq + \Pi_{59}$ with $A, B = N_{59} \pm 2\Pi_{59}$, i.e. the Pell–type system $Bx^2 - Ay^2 = -4\Pi_{59}^2$ with 113–digit coefficients. We verified that no accessible modulus obstructs it (mod 32, and all odd $\ell < 4000$: the conditions pin $q \equiv 5 \pmod 8$ and thin each $\ell \mid \Pi_{59}$ to about half of its residue classes and each $\ell \nmid \Pi_{59}$ to about a quarter, but never to zero), consistent with the local–solubility theme of Section 2. The obstruction, however, lives at the primes *of* A and B, and it is detectable *without factoring*: Proposition 1.2 below closes this family for *every* q by a single Jacobi–symbol computation. (An exact sweep to $q \leq 10^{12}$, `code/hunt/tail_sweep.rs`, 1.25×10^{11} candidates, none with $Aq + \Pi_{59}$ even a single

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square, which corroborates it independently.) The tail family also exposes the self–similarity of the ladder: writing $a = qm$, so that $m\,b = \Pi_{59}$ partitions the first 59 primes, its two closing conditions collapse to the exact equation

$$D(m)\,D(b) + m^2 \;=\; \Pi_{59}$$

together with the divisibility $m \mid D(b)$ (then $q = D(b)/m$, required prime); verified exhaustively on a 12–prime toy universe. Equivalently, the deficit $1 - \sigma(m)\sigma(b)$ must equal m^2/Π_{59} *exactly*: the circularity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$ of Section 5 of the core note in its purest form, with the deviation forced to be the perfect square m^2. Closing even this one family is thus a 2^{58}–fold exact–coincidence search of the same type as #307 itself (expected count $\sim 2^{58}/\Pi_{59} \approx 10^{-95}$): each rung of the ladder past 60 contains the whole problem. The one case with *no* free large prime (a 60–set drawn entirely from small primes) is instead a finite search, and empty in the minimal range: a direct both–squares enumeration of all 60–prime sets with $T > 2$ supported in the first 70 primes (4,011,289 sets; `code/hunt/close60.rs`) finds none passing even the plus–square, so no level–60 solution has all its primes ≤ 349. (The first 68 primes give 1,258,448 sets, also empty, and reproduce independently.) By Corollary 1.5 this verdict is decisive rather than provisional: at these masses a support passing the plus– and minus–squares, with the parity of Proposition 1.1(2), *is* a two–cycle, so a survivor would have been a solution and not a candidate. The enumeration extends with compute.

Proposition 1.2 (The canonical tail family is empty: a reciprocity kill)**.** *No solution of #307 has support $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$, for any prime q.*

Proof. A solution supported on U forces $x^2 = Aq + \Pi_{59}$ with $x = a + b$ and $A = N_{59} + 2\Pi_{59}$ (Remark 1.1). Every prime $\ell \mid A$ exceeds 277: for $\ell \leq 277$ one has $\ell \mid \Pi_{59}$, so $\ell \mid A$ would force $\ell \mid N_{59}$, contradicting $\gcd(N_{59}, \Pi_{59}) = 1$ (rigidity). Hence $\ell \nmid \Pi_{59}$, and reduction mod ℓ gives $x^2 \equiv \Pi_{59} \pmod{\ell}$, requiring the Legendre symbol $(\Pi_{59} \mid \ell) \neq -1$. But the Jacobi symbol satisfies

$$\left(\tfrac{\Pi_{59}}{A}\right) \;=\; -1$$

(a finite gcd–speed computation, script `code/tailkill.py`; no factorisation of the 114–digit A is needed), so some $\ell \mid A$ of odd multiplicity has $(\Pi_{59} \mid \ell) = -1$. The family is empty, for every q.

The companion symbol at $B = N_{59} - 2\Pi_{59}$ equals -1 as well; necessarily: for any admissible base S (writing $D = D_S$, $N = N_S$, $A_S, B_S = N \pm 2D$) one has $A_S \equiv B_S \pmod 8$ (as $4D \equiv 8 \bmod 16$) and $A_S \equiv B_S \equiv N \pmod p$ for every odd $p \mid D$, while the quadratic–reciprocity sign difference carries the exponent $\frac{p-1}{2} \cdot \frac{A_S - B_S}{2}$ with $\frac{A_S - B_S}{2} = 2D$ even; hence $(D \mid A_S) = (D \mid B_S)$ always; one binary certificate per tail family. Running it over all 49,961 admissible bases of Proposition 4.5 proves 31,219 of the one–new–prime families at level 60 empty (62.5%, curiously near 5/8); the remaining 18,742 are Jacobi–inconclusive: their -1–Legendre primes, if any, occur to an even count, visible only through a factorisation of A_S. A factoring attack on those (trial division and Brent–Pollard ρ over a Montgomery core, `code/hunt/survivor_kill.rs`; the found factors are kept as a pairwise–coprime basis, so a kill certificate (an odd–multiplicity basis piece P with $(D_S \mid P) = -1$) needs no primality testing at all), pushed to ρ–budget 10^6 with a Pollard $p - 1$ stage, closes a further 15,264: in total 46,483 of the 49,961 one–new–prime families at level 60 are proven empty (93.0%). The 3,478 still open (3,444 of them once Proposition 1.5 has emptied the 34 immune ones) resist all of this for a structural reason, not a want of effort: each is Baillie–PSW *composite* yet has no prime factor below $\sim 10^{12}$, so its A_S (or B_S) is a *balanced* semiprime of two $\sim$57–digit primes, splitting one is factoring at cryptographic scale. Two cheap tests confirm no shortcut survives: a product–tree batch–GCD over all 37,484 values A_S, B_S finds only shared primes below 10^9 (already found by ρ, so *no* free kills), and Baillie–PSW leaves exactly 34 families in which A_S and B_S

are both *probable* primes; these are immune *conditionally on that primality* (a prime A_S with $(D_S \mid A_S) = +1$ has no inert factor, hence no congruence obstruction for any q). The conditionality is not pedantry and not removable by more of the same computation: Baillie–PSW is rigorous only in its *composite* verdicts, so the 46,483 kills are unconditional while the 34 immunities are not. Upgrading them would require primality *certificates* for thirty-four 114–digit integers (ECPP or a Pocklington factorisation of A_S-1). The elementary route does not reach: immunity needs *both* A_S and B_S prime, and partial factorisation of $n-1$ (trial division to 3×10^6 together with the fully split part) leaves $\log F/\log n$ below $1/3$ on at least one side of every one of the 34, so neither Pocklington ($F > \sqrt{n}$) nor Brillhart–Lehmer–Selfridge ($F > n^{1/3}$) applies to any family: 0 of 34 are certifiable this way (`code/immune_certify.py`). This is now settled. Running the Adleman–Pomerance–Rumely –Cohen–Lenstra test on all 68 integers proves every one of them prime in 8 seconds of wall clock, so the 34 immunities upgrade from [C] to [P] and are *unconditional* (`code/immune_prove.gp`, PARI/GP 2.17.4, whose `isprime` is a proof and not a probable–prime test). The run reproduces the whole chain independently: 49,961 admissible bases, 31,219 Jacobi kills, and, among the 81 bases whose A_S and B_S are both prime, exactly the 34 with $(D_S \mid A_S) = +1$; the other 47 carry $(D_S \mid A_S) = -1$ and were already killed. The primality is moreover backed by *independently checkable* evidence rather than a verdict: `code/immune_certify.gp` emits Atkin–Morain ECPP certificates for all 68 integers, archived at `data/certs/immune_ecpp.txt` (512 KB) so a third party can check them without trusting the prover. That file is a `primecertexport` dump and is not machine–readable: PARI's `read` parses its header as a polynomial, so `primecertisvalid` cannot consume it. The working check is `code/immune_cert_verify.gp`, which extracts the 68 asserted subjects and re–proves each by ECPP in 1.3 seconds: 68/68 prime, 0 failures. The structural half of the campaign is now formal: the identity $(D_S \mid A_S) = (D_S \mid B_S)$, which is what makes one binary test decide a family rather than two, is proved in Lean 4 as `Erdos307.jacobiSym_A_eq_B` (`lean/Erdos307/Campaign.lean`), on the three standard axioms and with no `native_decide`; it follows from periodicity, $A_S - B_S = 4D_S$ being exactly the period of $J(a \mid \cdot)$. The primality itself is not formalised and cannot presently be: mathlib's only route to a large prime is `lucas_primality`, which needs a complete factorisation of $n-1$, and that is precisely what the 0 of 34 above rules out. Note the two counts answer different questions and both are correct: 81 is the number of prime pairs, 34 the number of *immune* families among them. A complete kill/immune split of the residual is in any case beyond feasible computation, and the honest state of the single–tail sector is 46,483 *empty* [P] *and* 3,478 *open*, the latter including the 34 immune ones, whose primality the APRCL run above makes unconditional [P]; Proposition 1.5 then empties those 34, leaving 3,444. □

Proposition 1.3 (Sector decomposition of level 60)**.** *Let U be a 60–element prime support with $T(U) > 2$. Then exactly one of the following holds:* (i) $T(U) - 2 \geq 1/\max U$*, and* $U = S \cup \{q\}$ *for an admissible base S (one of the* 49,961 *of Proposition 4.5); the* single–tail sector*, or* (ii) $T(U) - 2 < 1/\max U$*, and, writing $p > q$ for the two largest elements and W for the rest,* $T(W \cup \{p\}) < 2$ *and* $T(W \cup \{q\}) < 2$*; the* pair sector*, whose members escape every single–tail family.*

Proof. U lies in the tail family of $U \setminus \{r\}$ iff $T(U) - 1/r \geq 2$, which is easiest for $r = \max U$; (i) is that case ($T = 2$ being impossible by rigidity, the base is admissible). In case (ii) no removal reaches 2, and a fortiori the two stated 59–subsets stay below 2. □

Remark 1.4 (The campaign; completeness of the certificate; the arity barrier)**.** The reciprocity certificate captures *every* q–independent congruence obstruction of a tail family: at a prime $\ell \mid A_S B_S$ the system degenerates to the rank–one congruences $x^2 \equiv D_S$ or $y^2 \equiv D_S \pmod{\ell}$, while at $\ell \nmid 2A_S B_S$ the conditions define conics in (x, q) or (y, q) with $\sim \ell$ points, deleting at most half of the residue classes of q and never all (the 2–adic layer pins $q \equiv 5 \bmod 8$ without emptiness);

by CRT and Dirichlet, admissible q survive any finite set of such conditions. Hence a tail family admits a congruence kill iff $(D_S \mid \ell) = -1$ for some $\ell \mid A_S B_S$, which the Jacobi symbol detects for 31,219 bases, and partial factorisation (`code/hunt/survivor_kill.rs`) for a further 15,264: in total 46,483 of the 49,961 single–tail families at level 60 are proven empty (93.0%) and 3,478 remaining open. Of the latter, 34 have A_S and B_S both prime and are emptied by Proposition 1.5; the other 3,444 carry a balanced ∼57–digit semiprime and are factoring–hard. The 34 immune families are not a bottleneck; they are the tractable ones, and they are empty. Immunity means A_S is prime, and that is exactly what makes the family decidable directly.

Proposition 1.5 (Immune families are decided in polynomial time)**.** *Let S be a level–60 base with $A_S = N_S + 2D_S$ prime and $(D_S \mid A_S) = +1$. Then a solution in the tail family $S \cup \{q\}$ forces $x^2 = A_S q + D_S$ with x one of the two square roots of D_S modulo A_S, so the family has at most two candidate tail primes, computable by Tonelli–Shanks. All 34 immune families at level 60 are empty.*

Proof. Write $\sigma = N_S/D_S$, so $A_S = (\sigma + 2)D_S$. Proposition 4.7 bounds $q \le tD_S$ with $t = \frac{1}{2}(\sigma + \sqrt{\sigma^2 - 4})$, whence $x^2 = A_S q + D_S \le \big((\sigma + 2)t + 1\big)D_S^2$. At level 60 one has $\sigma > 2$ and σ close to 2, so $(\sigma + 2)t + 1 < 5.202$ and $x < 2.281\, D_S < (\sigma + 2)D_S = A_S$; over all 49,961 admissible bases the ratio x/A_S is at most 0.56982. Hence x lies in $[0, A_S)$ and, A_S being prime, is one of the two roots of $X^2 \equiv D_S$. Each root gives one candidate $q = (x^2 - D_S)/A_S$, to be tested for integrality, primality and the companion condition that $B_S q + D_S$ be a square. Carrying this out on the 34 families, whose A_S and B_S are unconditionally prime by the APRCL and ECPP runs recorded above, every candidate fails (`code/immune_decide.gp`), either because $q = (x^2 - D_S)/A_S$ is not a prime exceeding 787 or because $B_S q + D_S$ is not a square; the conclusion is therefore [P], not [C]. The script re–derives the 49,961 bases, the 31,219 Jacobi kills and the 34 immune families from scratch before deciding them. □

Primality of A_S plays no role in the bound $x < A_S$, only in the ease of extracting the roots, so the argument is general.

Proposition 1.6 (The split sieve)**.** *Let S be a finite set of primes, $D = \prod S$, and let M be a finite set of primes disjoint from S. Suppose the pair $a = \prod(T \cup M)$, $b = \prod(S \setminus T)$ with $T \subseteq S$ is a derivative two–cycle. Then*

$$\prod T \;\Big|\; \partial\big(\textstyle\prod(S \setminus T)\big), \qquad \Big(\prod T\Big)^2 \partial\big(\textstyle\prod M\big) + \partial\big(\textstyle\prod T\big)\, \partial\big(\textstyle\prod(S \setminus T)\big) = D.$$

The second identity carries no reference to the tail; the first is its reduction modulo $\prod T$, Theorem 2.3 removing the factor $\partial(\prod T)$, and is therefore a consequence of it rather than an additional hypothesis. Writing $\alpha = \prod T$ and $\beta = \prod(S \setminus T)$, the divisibility is by Lemma 2.1 the system of congruences $\sum_{p \in S \setminus T} p^{-1} \equiv 0 \pmod r$, one for each $r \in T$, so the expected number of surviving T is $\prod_{p \in S}(1 + 1/p)$ out of $2^{|S|}$. Neither statement involves A_S, so the criterion applies to the families whose A_S is composite, which Proposition 1.5 cannot reach, and to the pair sector of Proposition 1.3(ii), where $|M| = 2$ and $\partial(\prod M) = p + q$. Two consequences are recorded. Since $\partial(\alpha)\partial(\beta) > 0$ for T and $S \setminus T$ both nonempty, the identity forces $\alpha^2 < D$. And for X a set of odd primes $\partial(\prod X) \equiv |X| \pmod 2$, so $2 \in T$ forces $|T|$ odd. 0 *`sorry`* (`Erdos307.split_criterion_gen`, `split_snd_gen`, `split_snd_of_criterion`, `csum_odd_card`; `lean/Erdos307/SplitSieve.lean`).

Proof. Leibniz on the disjoint union gives $\partial(\prod(T \cup M)) = \prod M \cdot \partial(\prod T) + \prod T \cdot \partial(\prod M)$. The equation $\partial(b) = a$ reads $\partial(\beta) = \alpha \prod M$, which is the divisibility. The equation $\partial(a) = b$ reads

$\beta = \prod M \cdot \partial(\alpha) + \alpha\, \partial(\prod M)$; multiplying by α, replacing $\alpha \prod M$ by $\partial(\beta)$ and using $\alpha\beta = D$ gives the identity. For the converse direction of the first claim, reduce the identity modulo α: both $\alpha^2 \partial(\prod M)$ and D vanish, so $\alpha \mid \partial(\alpha)\partial(\beta)$, and $\gcd(\alpha, \partial(\alpha)) = 1$ by Theorem 2.3. $\square$

Remark 1.7 (What the sieve decides)**.** The criterion was applied to every level–60 single–tail base surviving the certificate of Proposition 1.2, 18,742 of them, a superset of the 3,444 open families, over all T with $|T| \le 6$: about 9×10^{11} splits, no factorisation, and no two–cycle. The observed survivor count is 5.20 per base against the predicted 5.17 (`code/split_sieve.rs`, `code/gen_bases.rs`). The range $|T| \ge 7$ is not decided. Its expected survivor count over $7 \le |T| \le 34$ is about 28, and deciding it by enumeration costs at least 7×10^2 core–years, so the sieve reduces the level–60 single–tail question without closing it.

Proposition 1.8 (Single–tail families are decided by factoring)**.** *Let S be a set of primes containing 2, with $D_S = \prod S$ squarefree, and let q be a prime completing S to a solution support. Then A_S is odd and coprime to D_S, and $x = a + b$ satisfies $x^2 = A_S q + D_S$ with $0 < x < A_S$. Hence x is one of the $2^{\omega(A_S)}$ square roots of D_S modulo A_S, each giving a single candidate $q = (x^2 - D_S)/A_S$, and the family is decided by one factorisation of A_S together with $O(2^{\omega(A_S)})$ primality tests.*

Proof. Since $2 \in S$ and D_S is squarefree, exactly one term of $N_S = \sum_{p \in S} D_S/p$ is odd, so N_S is odd and $A_S = N_S + 2D_S$ is odd. If $p \in S$ divided A_S it would divide N_S, but $N_S \equiv D_S/p \not\equiv 0 \pmod p$; as the primes of D_S are exactly S, $\gcd(A_S, D_S) = 1$. For the bound, write $\sigma = N_S/D_S$ and $A_S = (\sigma+2)D_S$. Proposition 4.7 gives $q \le tD_S$ with $t = \frac{1}{2}(\sigma + \sqrt{\sigma^2 - 4})$, so $x^2 \le \big((\sigma+2)t+1\big)D_S^2$. Across all 49,961 admissible level–60 bases $\sigma \in (2, 2.00236]$, whence $(\sigma+2)t + 1 < 5.202$ and $x < 2.281\, D_S$, while $A_S = (\sigma+2)D_S > 4\, D_S$; the ratio x/A_S is at most 0.56982 over the whole set (`code/lvl60check.gp`). Thus $x < A_S$. When instead $\sigma \le 2$, as for the pair–sector bases of Proposition 1.3(ii), Corollary 4.11 bounds the tail by $q \le D_S/(2D_S - N_S) = D_S/|B_S|$, so $x^2 = A_S q + D_S \le A_S D_S/|B_S| + D_S < A_S^2$ because $|B_S| \ge 1$ and $A_S > D_S$; the conclusion $x < A_S$ therefore holds in both regimes. Being odd and coprime to D_S, A_S admits exactly $2^{\omega(A_S)}$ square roots of D_S when D_S is a residue modulo every prime power dividing it, and none otherwise; these are computed from the factorisation by Tonelli–Shanks, Hensel lifting at any repeated factor, and the Chinese remainder theorem. $\square$

This replaces a 2^{58} search per family by a single factorisation, and the integer to be factored is $A_S \approx 4D_S$, of 114 decimal digits, not the 230–digit product the divisor search ranges over. It also supersedes the reciprocity route rather than extending it: a congruence kill decides at best half of the split families, since for $A = pq$ with $(D_S \mid A) = +1$ the two Legendre symbols are either both $+1$, when no kill exists, or both -1, when one does, whereas the root enumeration decides every family whose A_S is factored. One cheap idea deserves to be recorded as closed, because the level–60 bases are far from random: all 49,961 share the 39 forced primes below 167, so one may hope that two of the A_S share a prime factor and split each other for free. A batch GCD in the sense of Bernstein (one product tree over the hard cofactors, then a remainder tree) tests this for every pair at once in seconds. Over the 7,713 families that survive the Jacobi certificate and trial division to 10^6, a superset of the 3,478, exactly two of the 6,972 composite cofactors of A_S share a factor, and both shared factors have seven digits, so they lie barely past the trial–division bound and are found instantly by any factoring method. The structure the bases share is in D_S and does not propagate to $A_S = N_S + 2D_S$: at the wall the residual behaves as a set of pairwise coprime balanced semiprimes, and no amount of shared structure among the bases is going to split them. (`code/batchgcd.gp`)

Level 60 is therefore not an exponential obstruction but a factoring project: the 34 with A_S prime are already done, and 3,444 integers of 114 digits remain. A 114–digit general number field sieve is routine, costing core–days apiece rather than the 3.2×10^7 core–years the divisor search of Proposition 4.7 would demand, a reduction by six orders of magnitude. It should be said plainly that this does not by itself deliver $|P \cup Q| \geq 61$. The single–tail sector is only one of the two cases of Proposition 1.3, and the pair sector contributes a further 3,013,379 surviving bases, each an arity–one tail family to which Proposition 1.8 applies equally. Factoring those as well is a computation some three orders of magnitude larger, tens of thousands of core–years, which is beyond what has been carried out here and is quoted as the honest price, not as a claim of feasibility. What has changed is the character of the obstruction: level 60 is no longer exponential in $|S|$, but linear in the number of families, each at the cost of one factorisation.

Those 3,478 are expected to be empty, and by a wide margin: evaluating the expected count of Remark 1.13 over them, with $\kappa_S \leq 1$ and the tail sum bounded by $\log\log(tD_S)$, gives at most 5.5×10^{-108}, the bases having D_S of 113 digits. The divergence of the global count comes from large a and not from this level, and the first solution the heuristic locates sits at $10^{116.7}$ or beyond. So the factoring wall guards the verification of something the heuristic already asserts, rather than a plausible solution. The weapon is intrinsically *arity–one*, and provably fails on the pair sector. With $s = p + q$, $t = pq$, a pair–sector member requires the two simultaneous squares $x^2 = A_W t + D_W s$ and $y^2 = B_W t + D_W s$ (the would–be third condition $s^2 - 4t = \square$ is automatic, being $(p - q)^2$), together with the reciprocity–linked constraints $(D_W p \mid q) = (D_W q \mid p) = 1$ (mod q, a solution with $q \mid a$ has $N_U \equiv D_W p$, forcing $D_W p$ to be a square). At a prime $\ell \mid A_W$ the condition $x^2 \equiv D_W s$ (mod ℓ) constrains only s, which the free second prime absorbs; the pair system is in fact locally soluble at *every* modulus (a theorem, Proposition 1.9 below, corroborated by a direct scan over all $\ell \leq 600$) so, unlike the single–tail family, the pair sector admits *no* congruence obstruction at all. It is a genuine two–parameter surface per base, #307 in miniature (Remark 1.12); reciprocity cannot touch it, and only a direct both–squares search or a descent argument can.

The sector is nevertheless an explicit finite object, and small. A level–60 support U is covered by the one–new–prime campaign exactly when $T(U \setminus \{\max U\}) > 2$, the bases of Proposition 4.5 being the 59–prime sets of mass exceeding 2; the complement is $U = R \cup \{m\}$ with $|R| = 59$, $m = \max U$ and $T(R) \leq 2 < T(R) + 1/m$. The mass ladder at $k = 60$ puts every element of R below $793.67 \times 2 = 1587.4$, so the complement is a search over 59–subsets of the 250 primes under 1588 with $T(R) \in (2 - 1/\max R,\, 2]$. There are exactly 18,234,653 of them, the largest $\max R$ being 1583 (`code/pairsector_count.rs`, corroborated by an independent binned counting DP in `code/pairsector_countdp.rs`). Each is an arity–*one* tail family in m: Proposition 1.9 denies a congruence kill to the family with both new primes free, not to these, so the q–independent certificate does apply to each of the 18.2 million bases. The immunity of the pair sector is a consequence of its two free parameters, and fixing one of them buys the weapon back.

That the sector is finite at all rests on $T(R) \neq 2$: at $T(R) = 2$ every m would satisfy $T(R) + 1/m > 2$ and the tail would be unbounded. It cannot occur. If $2 \notin R$ then D is odd and each D/p is odd, so $N = \sum_{p \mid D} D/p$ is a sum of 59 odd terms and is odd; if $2 \in R$ then $D/2$ is odd and the other 58 cofactors are even, so N is again odd. Since $2D$ is even, $N \neq 2D$ (`Erdos307.mass_ne_two`; `code/pairsector_parity.gp`).

Running the certificate over the sector kills 10,457,149 of the 18,234,653 bases (57.3%) by the Jacobi symbol alone, and trial division of A_R and B_R to 10^6 closes a further 4,580,243 of the survivors (58.9% of them): in total 15,037,392, or 82.5%, with 3,197,261 still open (`code/pairsector_kill.rs`, `code/pairsector_factor.rs`; the same code in campaign mode reproduces the 49,961, 31,219 and 18,742 of the paragraph above exactly). A Pollard rho pass with a budget of 5×10^4 iterations per base then kills a further 183,882, leaving 3,013,379 open (83.5% of the sector eliminated); the marginal return has fallen to 5.7%, which is what one expects once the small factors are gone and what remains is a balanced semiprime. The pair sector is therefore not a barrier of a different kind from the one–new–prime families: it is the same computation at 365 times the scale, and it responds to the same weapon.

Proposition 1.9 (The pair sector admits no congruence kill)**.** *Let W be any pair–sector base and $A = A_W$, $B = B_W$, $D = D_W$, so that a pair–sector solution requires $x^2 = A\,\alpha\beta + D(\alpha+\beta)$ and $y^2 = B\,\alpha\beta + D(\alpha+\beta)$ with α, β the two new primes. For every prime power ℓ^j this system has a solution in units α, β and integers x, y at a nonsingular point. Consequently, by CRT and Dirichlet, no finite system of congruence conditions empties any pair–sector family: the arity barrier of Remark 1.4 is a theorem, and the certificate programme provably ends at arity one.*

Formalised, in part. *`Erdos307.no_pinning_isUnit` is the step that makes the tail–kill mechanism unavailable, `collapse_identity` is regime* (2)*'s observation that the two square conditions are one, `regime_one_first` and `regime_one_second` are the explicit unit of regime* (1)*, and `targets_agree_mod_eight` is the mod–8 collapse of regime* (3)*. The lift from ℓ to ℓ^j that regime* (1) *needs is `square_lifts_to_prime_powers`, proved from `hensel_all_powers`: if ℓ is an odd prime not dividing c and c is a square modulo ℓ, it is a square modulo every power of ℓ. Regime* (2) *carried a citation to Weil's bound in earlier revisions; it does not need one. Its two steps are `Erdos307.pairlocal_inner_sum`, the evaluation of the inner sum of S_{fg} as $-\chi\big((A\alpha+D)(B\alpha+D)\big)$ by the quadratic complete sum of Proposition 1.14, with its side condition discharged as $4D^2\alpha^2 \neq 0$; and `pairlocal_count_pos`, the arithmetic turning the resulting $O(\ell)$ bounds into a positive count from $\ell \geq 17$.* 0 *`sorry`* (`lean/Erdos307/PairLocal.lean`, `lean/Erdos307/PairLocalCount.lean`)*. Checked directly: over* 872 *random bases with $A - B = 4D$ and every odd $\ell \leq 400$, the count of admissible (α, β) was positive in every case and tracks $(\ell-1)^2/4$; the crude bounds $|S_f|, |S_g| \leq 2\ell$, $|S_{fg}| \leq 3\ell$ held in all* 409 *cases tested* (`code/pairlocal_count.gp`)*.*

Proof. First, no ℓ pins either target to a constant: that needs $\ell \mid A$ (or B) *and* $\ell \mid D$, whence $\ell \mid N$, contradicting $\gcd(N, D) = 1$ (rigidity); so the tail–kill mechanism is unavailable, and solubility is decided in three regimes.

(1) ℓ odd, $\ell \mid D$. This covers every odd $\ell \leq 103$: at level 60, omitting a prime p caps $T(U)$ at $T_{61} - 1/p$, and $1/p > T_{61} - 2 = 0.00944\ldots$ exactly for $p \leq 103$, so all such primes lie in U, hence in W (the two adjoined primes are the largest). Take $\beta = 1$, $x = 1$, $\alpha = (1-D)/(A+D) \bmod \ell^j$, a unit since $A + D \equiv N \not\equiv 0 \pmod{\ell}$: the first equation holds by construction, and the second target $(B+D)\alpha + D \equiv N \cdot N^{-1} = 1 \pmod{\ell}$ is a unit congruent to a square, so y exists modulo ℓ^j.

(2) ℓ odd, $\ell \nmid 2D$ (in particular every $\ell \geq 107$ outside W). The two conditions are secretly *one*: setting $m = (x^2 - y^2)/4D$ and $u = (x^2 - Am)/D$, the relation $A - B = 4D$ gives the polynomial identity $Bm + Du = y^2$. Only *existence* is wanted, and counting the pairs (α, β) directly settles it with no curve and no Weil bound. Write χ for the quadratic character, $f = A\alpha\beta + D(\alpha+\beta)$ and $g = B\alpha\beta + D(\alpha+\beta)$. Summing $\frac{1}{4}(1+\chi(f))(1+\chi(g))$ over $\alpha, \beta \in \mathbb{F}_\ell^\times$ and discarding the at most $2(\ell-1)$ pairs with $fg = 0$,

$$4\,\#\{\text{both nonzero squares}\} \;\geq\; (\ell-1)^2 + S_f + S_g + S_{fg} - 8(\ell-1).$$

For fixed α each target is *linear* in β, so the inner sums of S_f and S_g vanish by the linear evaluation above, leaving $|S_f|, |S_g| \leq 2\ell$. And fg is a *product of two linear forms* in β, so its inner sum is the quadratic evaluation, whose side condition $ad \neq bc$ is here $D\alpha(A-B)\alpha = 4D^2\alpha^2 \neq 0$, exactly the hypotheses $\ell \nmid 2D$ and $\alpha \neq 0$; summing the resulting $-\chi\big((A\alpha+D)(B\alpha+D)\big)$ over α is that same evaluation again, so $|S_{fg}| \leq 3\ell$. Hence the count is positive as soon as $(\ell-1)^2 - 7\ell - 8(\ell-1) > 0$, i.e. from $\ell \geq 17$, far below the $\ell \geq 107$ in play. The point is nonsingular ($\alpha\beta \neq 0$; the Jacobian contains $\mathrm{diag}(2x, 2y)$), so it lifts to every ℓ^j.

(3) $\ell = 2$. Both targets are odd and agree modulo 8 (as $\alpha\beta$ is odd, $D(\alpha+\beta) \equiv 0 \bmod 4$, and $(A-B)\alpha\beta = 4D\alpha\beta \equiv 0 \bmod 8$), and a finite table over ($A \bmod 8$, $D \bmod 8$) shows that for every

base some achievable pair $(\alpha\beta, \alpha+\beta) \bmod 8$ makes the common target $\equiv 1 \pmod 8$; a 2–adic square, soluble modulo 2^j for every j.

All three regimes, the collapse identity, the threshold $1/103 > T_{61} - 2 > 1/107$, and the mod–32 layer were verified computationally on three bases and every $\ell \le 600$ (`code/pairlocal.py`; the measured density of admissible (x, y) in regime (2) is 0.497–0.503 against the predicted $\frac{1}{2}$). $\square$

Remark 1.10 (Anatomy of the certificate; the empirical 5/8 law)**.** Writing $D = 2D_1$, the reciprocity signs in the certificate cancel ($\frac{A-1}{2} + \frac{N-1}{2} = N - 1 + 2D_1$ is even), leaving the clean product

$$\left(\tfrac{D}{A}\right) = \left(\tfrac{2}{A}\right)\left(\tfrac{D_1}{N}\right), \qquad \left(\tfrac{2}{A}\right) = +1 \iff N \equiv 3, 5 \pmod 8,$$

and one layer down $(N \mid D_1) = \prod_{p \mid D_1} \left(\frac{D/p}{p}\right)$, since $N \equiv D/p \pmod p$: the cofactor structure of rigidity recurs inside its own certificate. This product evaluates in closed form: with $D/p = 2D_1/p$ and quadratic reciprocity,

$$(N \mid D_1) = (2 \mid D_1) \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{s + \binom{t}{2}}, \quad s = \#\{p \mid D_1 : p \equiv 3, 5\ (8)\},\ \ t = \#\{p \mid D_1 : p \equiv 3\ (4)\},$$

a Rédei genus character of D_1, *independent of* N. This is forced, not merely observed: $(N \mid D_1) = \prod_{p \mid D_1} (N \mid p)$, and cofactor survival gives $N \equiv D/p = 2D_1/p \pmod p$, so $(N \mid p) = (2 \mid p)\,(D_1/p \mid p)$. The first factors multiply to $(-1)^s$ since $(2 \mid p) = -1$ exactly for $p \equiv 3, 5 \pmod 8$; and, each unordered pair $\{p, q\}$ occurring once in each of the two cofactors,

$$\prod_{p \mid D_1} \left(\frac{D_1/p}{p}\right) = \prod_{\{p,q\} \subset D_1} \left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = \prod_{\{p,q\} \subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{\binom{t}{2}},$$

by quadratic reciprocity, since $\frac{p-1}{2}$ is odd exactly for $p \equiv 3 \pmod 4$. (Checked independently, prime by prime, on 20,000 bases: 0 violations.) The inner symbol is thus not a coin but a genus invariant; its empirical fairness (49.9%, 50.0% over the 49,961 bases) is that character's equidistribution; and $(D_1 \mid N)$ follows by the reciprocity sign $(-1)^{\frac{D_1-1}{2}\frac{N-1}{2}}$, hence depends on N only through $N \bmod 4$. In fact the entire certificate closes into a finite functional. Let $v(S) = (n_1, n_3, n_5, n_7) \bmod 4$ count the odd primes of S in the residue classes $1, 3, 5, 7$ modulo 8. Every factor above is a function of v alone: $s \equiv n_3 + n_5$, $t = n_3 + n_7$ (with $\binom{t}{2}$ needing only $t \bmod 4$), $D_1 \equiv 3^{n_3} 5^{n_5} 7^{n_7} \pmod 8$, $S(D_1) \equiv (n_1 + n_5) - (n_3 + n_7) \pmod 4$, and (by the mod–8 law of Proposition 1.14, $D_1' \equiv D_1 S(D_1) \pmod 8$) also $N = D_1 + 2D_1' \equiv D_1 + 2\big(D_1 S(D_1) \bmod 4\big) \pmod 8$. Hence

$$(D \mid A) = K\big(v(S)\big):$$

the kill is decided by sixteen residue counts modulo 4. The realized vectors are exactly the coset $n_1 + n_3 + n_5 + n_7 \equiv 58 \equiv 2 \pmod 4$ (all 64 of its classes occur), K is constant on each, and the functional reproduces the direct 113–digit Jacobi computation on every one of the 49,961 bases (`code/kill_classvector.py`). The five–eighths law is then an exact finite statement, and its mechanism is the *reciprocity switch*: for $N \equiv 1 \pmod 4$ the reciprocity sign is inert and the kill reduces to the genus coin; exact conditional rate $12{,}699/24{,}975 = 0.5085$; while for $N \equiv 3 \pmod 4$ the sign couples the parity of t into the product and the rate jumps to $18{,}520/24{,}986 = 0.7412$ (by class modulo 8: 0.509, 0.741, 0.508, 0.741 at $N \equiv 1, 3, 5, 7$; the 3/4 classes are $N \equiv 3 \pmod 4$, not the $(2 \mid A)$ classes). The blend $\frac{1}{2} \cdot \frac{24975}{49961} + \frac{3}{4} \cdot \frac{24986}{49961} = 0.62503$ against the exact $31{,}219/49{,}961 = 0.62487$ is the whole of the "coincidence". The correlation is an ensemble property, a random–subset proxy gives ≈0.45/0.55; and the functional yields no second certificate: it recomputes the same symbol.

Remark 1.11 (Why a one–sided mass argument cannot reach the barrier)**.** The threshold $\sigma(U) = t + 1/t$ of Proposition 1.15 makes the barrier equivalent to a lower bound $|\sigma(a) - 1| > c$ with c absolute, and it is worth recording what such a bound cannot be proved from. The primary pseudoperfect numbers satisfy the one–sided equation $n' = n - 1$ exactly, so along them $|\sigma(n) - 1| = 1/n$ tends to 0: the eight known members give $1/n$ from $\frac{1}{2}$ down to 1.18×10^{-31}. They are solutions of the coprime relaxation (Corollary 8.7, the case $u = 1$). Hence no argument that uses only $a' = b$ with b squarefree and coprime to a can bound $|\sigma(a) - 1|$ below by an absolute constant, since the same argument would prove the corresponding statement for the relaxation, which is false. Any such bound must consume the second equation $b' = a$. This is why the mass arguments of Section 4 of the core note, and the Sylvester–type extremal bounds, reproduce 60 and stop: they are one–sided, and $1/a$ is the most a one–sided argument can give.

Remark 1.12 (The kill–program is one–sided)**.** Every kill is a proof that no solution hides in that family, but if #307 has a solution, the solution's own family is unkillable, so no accumulation of kills can overshoot the truth: the program's asymptote *is* the existence question. Together with the self–similarity of Remark 1.1 (each rung past 60 contains the whole problem), this locates the campaign precisely: it corners where a solution could live, one certified region at a time, and can terminate only if #307 has answer NO, which the heuristics disfavour.

Remark 1.13 (The reciprocity kills are the local factors of the expected count)**.** The divergent naive count of Section 2 and the reciprocity sieve of Proposition 1.2 are the *same* computation. Decompose the expected number of solutions over tail families $S \cup \{q\}$: the summand for a base S is $\kappa_S \left(D_S\sqrt{\text{mass}^2 - 4}\right)^{-1} \sum_q q^{-1}$, where $\kappa_S = \prod_\ell$(local solubility density at ℓ) is exactly the reciprocity data. A reciprocity–killed base carries an inert $\ell \mid A_S$ with $(D_S \mid \ell) = -1$, so $N_S + 2D_S \equiv D_S \pmod{\ell}$ is a non–residue for *every* q, forcing $\kappa_S = 0$; the family drops from the count entirely. An immune base (A_S, B_S both prime, both residues) has no inert prime, so $\kappa_S > 0$ and, treating q as ranging freely with density $1/q$, the summand *looks* like a divergent $\sum_q q^{-1}$ (verified on an explicit 114–digit immune base: local density 1 at every tested prime, partial sums growing without bound). That model overstates the space: Proposition 4.7 bounds q itself, so the true sum is finite, and Proposition 1.8 identifies it exactly with the roots of D_S modulo A_S, of which there are only two once A_S is prime. The immune part of the divergence was therefore never genuine, only an artifact of summing a local density over an unbounded range that Proposition 4.7 had already closed; the kills remove ∼93% of the count's mass, and Proposition 1.5 removes the rest of the 34 conditionally immune families' contribution unconditionally, both candidate values of q failing outright. What Remark 1.13 originally located as the entire "weakly favours YES" signal at level 60 is now zero there. What remains is the pair sector, and there the picture does not simplify the same way: the object Proposition 1.9 proves permanently immune to any finite congruence condition is the one with *both* new tail primes free, which the campaign never enumerates as such, fixing one of the two, as Remark 1.4 does to get a finite, countable search, buys the congruence weapon back for the resulting 18,234,653 bases, and their 83.5% elimination is a plain factoring–budget shortfall, the same story as the single–tail sector's, not an instance of Proposition 1.9's theorem. Proposition 1.8 applies to that computed population exactly as it does to the single–tail residual (its proof addresses the $\sigma \leq 2$ case via Corollary 4.11), so the residual open question there is, again, the single task of factoring a ∼114–digit integer per family. Proposition 1.9's genuinely uncomputed object is a separate matter: it is why the campaign stops at fixing one tail prime rather than a limitation discovered by trying and failing to go further. This does not touch the reliability of the count (which ignores the $N(r) \leq 1$ rigidity), but it retires the level–60 half of what it once pinned down as the heuristic's YES.

Proposition 1.14 (Local completeness of the reciprocity sieve)**.** *For an admissible base S and an odd prime ℓ, the admissible tail residues $Q_\ell = \{q \bmod \ell : (A_S q + D_S \mid \ell) \neq -1$ and $(B_S q + D_S \mid \ell) \neq -1\}$ fall into exactly three regimes.* (i) *$\ell \mid D_S$, which covers every $\ell \leq 167$ uniformly over the* 49,961 *admissible bases, since $1/p \geq (T_{59}-2)+\frac{1}{281}$ forces every prime $p \leq 167$ into every base: then $A_S \equiv B_S \equiv N_S \not\equiv 0 \pmod{\ell}$, the two conditions collapse to the single condition $(N_S q \mid \ell) \neq -1$, and $|Q_\ell| = (\ell+1)/2 \ni 0$.* (ii) *$\ell \mid A_S B_S$: one condition is the constant $(D_S \mid \ell)$, so $Q_\ell = \emptyset$ iff $(D_S \mid \ell) = -1$ (exactly the certificate of Proposition 1.2) and otherwise $|Q_\ell| \geq (\ell-1)/2$.* (iii) *$\ell \nmid A_S B_S D_S$ (hence $\ell \geq 173$): the complete character sums give $|Q_\ell| \geq (\ell-3)/4 - 2 \geq 40$ (the linear sums vanish, the quadratic sum is $-\chi(A_S B_S)$, boundary terms ≤ 2).* (*Regimes* (i) *and* (ii) *are formalised: `Erdos307.regime_one_collapse`, `regime_one_same_argument` and `regime_two_constant` (`lean/Erdos307/LocalComplete.lean`). Regime* (iii)*'s counting step is `regime_three_count`, and its two complete character–sum hypotheses are now theorems rather than citations: $\sum_q \chi(aq+b) = 0$ for $a \neq 0$ is `Erdos307.sum_quadraticChar_affine`, the reindexing $q \mapsto aq+b$ followed by `quadraticChar_sum_zero`; and $\sum_q \chi((aq+b)(cq+d)) = -\chi(ac)$ for $ad \neq bc$ is `Erdos307.sum_quadraticChar_quad` proved by factoring out $\chi(ac)$, translating to $\sum_u \chi\big(u(u+e)\big)$ with $e \neq 0$, writing $u(u+e) = u^2(1+e/u)$ for $u \neq 0$ so that the character value is $\chi(1+e/u)$, and reindexing by the involution $u \mapsto e/u$ of the nonzero elements to reach $\sum_{t \neq 0} \chi(1+t) = 0 - \chi(1) = -1$. Both hold over any finite field of odd characteristic.* 0 *`sorry`* (`lean/Erdos307/CompleteSums.lean`)*. Checked against direct evaluation for all odd primes $\ell \leq 59$: the linear sum over every (a,b) with $a \neq 0$, and the quadratic sum over all* 33,008,952 *quadruples with $ad \neq bc$, 0 violations* (`code/complete_sums_check.gp`).) *Consequently $Q_\ell \neq \emptyset$ forces $|Q_\ell| \geq 2$ at every modulus, the surviving local density is positive (e.g. $\prod_{\ell \leq 300} |Q_\ell|/\ell \approx 2.2 \times 10^{-19} > 0$ on the canonical base), and by CRT* no finite system of congruences (no covering) can annihilate an unkilled family*: the binary certificate of Proposition 1.2 is the complete local theory of the tail sector, there is no "second kill layer," and the one–sidedness of Remark 1.12 is structural rather than an artefact of method. (All three regimes verified exhaustively on the canonical base: collapse and $|Q_\ell| = (\ell+1)/2$ at $\ell \in \{3,5,7,101,277\}$; the Weil floor at every prime $281 \leq \ell \leq 600$; A_S, B_S carry no factor below 10^6, consistent with the balanced–semiprime wall of Remark 1.4.)*

Proposition 1.15 (What a barrier improvement must supply)**.** *Let (a,b) be a solution with $a < b$ and put $t = b/a$, so that $t = \sigma(a)$ and, by $\sigma(a)\sigma(b) = 1$,*

$$\sigma(U) \;=\; \sigma(a) + \sigma(b) \;=\; t + \tfrac{1}{t}.$$

For n with $T_n > 2$ let $t_n > 1$ be the root of $t + 1/t = T_n$. Since $\sigma(U) \leq T_{|U|}$,

$$\sigma(a) > t_n \implies |U| \;\geq\; n+1,$$

and this is the only route by which a mass argument can raise the barrier. Numerically

n	59	60	61	62	63	64
T_n	2.0023502	2.0059089	2.0094424	2.0128554	2.0161127	2.0193282
t_n	1.0496677	1.0798804	1.1020081	1.1199914	1.1352477	1.1490254

So the barrier level is decided entirely by how close $\sigma(a)$ may lie to 1, and each further level demands a definite gap: $|U| \geq 61$ needs $\sigma(a) > 1.0799$, a full 8% above 1.

That is why the mass mechanism is exhausted here rather than merely difficult. The only unconditional lower bound on the gap is $\sigma(a) - 1 = (a' - a)/a \geq 1/a$, which is worthless at $a > 10^{56}$; and σ genuinely comes close to 1 from above on very few primes, the Sylvester-type set $\{2,3,7,41\}$ giving $\sigma = 1.00058\ldots$ with four. Indeed no bound in $\omega(a)$ alone can exist: the greedy Sylvester construction gives, for each k, squarefree a with $\omega(a) = k$ and $0 < |\sigma(a) - 1|$ doubly exponentially small in k, from below $1 - \sigma = 7.8 \times 10^{-592}$ at $k = 12$, and from above $\sigma - 1 = 4.6 \times 10^{-591}$ at the same k (`code/sigma_gap.gp`). So $|\sigma(a) - 1| \geq f(\omega(a))$ is false for every positive f, and the unconditional $1/a$ is of the right shape. Nothing in the cycle equations forbids $a' - a = 1$. A barrier past 60 must therefore come from a mechanism that bounds $\sigma(a)$ away from 1 without appealing to the size of U, which is what Theorem 4.3 does in the opposite direction, and we know of none.

The same reduction locates the extreme case. Writing $c = b - a$, the identity $\sigma(U) - 2 = (t-1)^2/t = c^2/D$ is the minus layer $D' - 2D = c^2$, and $\gcd(c, D) = \gcd(b - a, ab) = 1$ because $\gcd(a, b) = 1$. Since an admissible support contains every prime ≤ 167 (Remark 1.1), c is divisible by none of them, so $c = 1$ or $c \geq 173$. The branch $c = 1$ is $D' = 2D + 1$, equivalently $a' = a + 1$ and $(a + 1)' = a$; it is the closest a solution could come to $\sigma(a) = 1$, and it is exactly the configuration a mass argument cannot reach. Below 3×10^6 the equation $a' = a + 1$ has only the solutions 30, 858, 1722 and 66,198, and none extends: for the first three $a + 1$ is prime, so $(a + 1)' = 1$, and for the last $a + 1$ is not squarefree. The branch is not excluded, only unsearched, since the barrier puts it above 10^{56}. The level–59 computation that gives $|U| \geq 60$ is, in these terms, an exhaustive verification that $\sigma(a) > t_{59}$, and Proposition 4.8 says the same verification is unavailable one level up.

Remark 1.16 (The tail is not a Lehmer question)**.** Remark 1.26 below reads the surviving question as the primality of a term q_n of the *exponential* Pell orbit, and Remark 2.26 names that orbit as the only live branch. Proposition 4.7 withdraws the reading. The orbit is genuinely infinite and both square conditions do hold along all of it; what fails is that no term past N can be prime. The single identity responsible is $A - B = 4D$, which the earlier treatment used only to form the Pell system and never to factor $4Dq$.

Remark 1.17 (The threshold in largest–prime coordinates, and which sectors it leaves alive)**.** Remark 1.1 already writes a cycle as $a = qm$ with q the largest prime; Bado [28] makes the same reduction exact and independent of the level. With $q = P^+(ab)$, $q \mid a$, write $a = qd$, $b = e$: then $e = d + qd'$, $e' = qd$, and $q = (e - d)/d'$. Here $\sigma(a) = 1/q + \sigma(d)$, so t (or $1/t$, according to which side carries q; $t + 1/t$ is symmetric and the reading below is unaffected) is $\sigma(d)$ up to $1/q$, and Proposition 1.15 becomes a statement about the *sector* d. It is very uneven. (The rows below take q astronomically large, which Proposition 4.5 supplies once d is small: at $q = 281$ the first row would read 1295.)

d	$\sigma(d)$	$\sigma + 1/\sigma$	$\lvert U\rvert \geq$
2	0.500000	2.500000	1413
$2 \cdot 3$	0.833333	2.033333	69
$2 \cdot 3 \cdot 7$	0.976190	2.000581	59
$2 \cdot 3 \cdot 7 \cdot 43$	0.999446	2.000000	59
$2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$	0.999979	2.000000	59
3	0.333333	3.333333	$\sim 1.1 \times 10^8$

The last row is a Mertens estimate, the exact K being past $\pi(10^6)$. Inverting $t + 1/t \leq T_K$ in the same convention gives the windows: $|U| \leq 60$ forces $\sigma(d) \in [0.9260, 1.0799]$, the t_{60} of the table above, and $|U| \leq 68$ forces $\sigma(d) \in [0.8372, 1.1945]$ (`code/sector_window.gp`). So a sector can be cheap only if $\sigma(d)$ sits within a few per cent of 1. The primary pseudoperfect numbers, having $\sigma(n) = 1 - 1/n$ exactly, are among those extremal sectors (they are not the only ones, the

Sylvester–type $\{2,3,7,41\}$ above being another) and by Corollary 8.7 they are the $u=1$ solutions of #307's coprime relaxation, so Erdős #313 sits at the floor of this barrier as well.

The reduction is not new to the tail theory: it *is* the parametrisation of Proposition 4.7, with $\alpha = d$ and $\beta = e$. Substituting $e = d + qd'$ and $e' = qd$ into $\alpha^2 q^2 - Nq + (\beta^2 - D)$ at $\alpha = d$ gives $e(e-d-qd') = 0$ identically (`code/alpha_is_d.gp`; the discriminant identity is $4d^2$ times the same vanishing, not a second fact). Two things follow. Everything proved about tail families is a statement about sectors: the reciprocity kill of Proposition 1.2, and the factoring wall behind the families it leaves open. And the sector $d = 2$ is $\alpha = 2$, which the table puts at $|U| \geq 1413$, so the level–60 programme never meets it. Negatively, the reduction does not shorten that residual: the unknown is the divisor $\alpha \mid D$, of which there are 2^{59}, and the only α–independent certificate is the Jacobi one already used.

Remark 1.18 (The sector barrier sharpened by d', and a second derivation of 60)**.** The sector reading above uses only the mass. It can be sharpened, and the sharpening is what Bado [6] exploits at $d = 42$. Since $e = d + qd'$, one has $\gcd(e,d) = \gcd(qd',d) = 1$ and $\gcd(e,d') = \gcd(d,d') = 1$, so

$$\operatorname{supp}(e) \text{ avoids } \operatorname{supp}(d) \cup \operatorname{supp}(d') \qquad \text{in every sector,}$$

and the mass $\sigma(e)$, which is $1/\sigma(d)$ to many places, must be carried by primes drawn from a thinned set. Two further identities come with it, both immediate from $e' = qd$: $de - d'e' = d^2$ and $\sum_{r\mid e} 1/r = d/d' - d^2/(d'e)$; and the associated defect $\Delta_d(R) = dR - d'R'$ satisfies $\Delta_d(Rr) = r\Delta_d(R) - d'R$ with terminal value $\Delta_d(e) = d^2$ (`code/sector_general.gp`, symbolic in d, d', q, R, R'). When d is primary pseudoperfect, $d' = d - 1$; that is the only reason those sectors look distinguished.

Running the resulting bound sector by sector (mass d/d' from primes avoiding $\operatorname{supp}(d) \cup \operatorname{supp}(d')$, together with the parity constraint that d even forces e odd and $\omega(e)$ even) gives

$$\min_d \bigl(1 + \omega(d) + \omega(e)\bigr) \;=\; 60,$$

over all sectors d built from primes up to 47, attained at $d = 15470 = 2 \cdot 5 \cdot 7 \cdot 13 \cdot 17$ and at many others, among them the primary pseudoperfect $47058 = 2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$ (`code/sector_dprime.gp`). This is Theorem 4.3's constant recovered by a route sharing nothing with its proof: that one is an exhaustive computation over the 49,961 admissible 59–prime supports, this one a largest–prime descent and a reciprocal–mass count. It does not improve the barrier, it bottoms out at exactly 60, but two independent derivations of the same constant are worth recording, and the sectors where it is tight are where any improvement must act.

At $d = 42$ the mass bound alone gives $\omega(e) \geq 62$ and $|U| \geq 66$; [6] obtains $\omega(e) \geq 64$ and $|U| \geq 68$ by adjoining the residue conditions $\sum_{r\mid e} r^{-1} \equiv 0 \pmod 3$ and $\pmod 7$ and $\prod_{r\mid e} r \equiv 1 \pmod{41}$ and maximising over admissible supports. We have re-run that certificate independently, obtaining the same maximum and the same 51,281 reachable states. The residue sieve cannot be pushed further as a sieve: the complete system of local conditions is the sector equation itself. For each $r \mid e$, reducing $e' = qd$ modulo r gives $\prod_{s\mid e,\, s\neq r} s \equiv -d^2/d' \pmod r$, one condition per prime of e; conversely these, with $e \equiv d \pmod{d'}$ and $1 < \sigma(e) < 2$, force $e' = qd$ exactly, both sides lying in $(e, 2e)$. Bado's conditions modulo 3 and 7 are the sums of these; the maximiser certifying his bound violates them at five of its six smallest primes, and adjoining the three at $5, 11, 13$ to the dynamic programme lowers the 64–prime maximum from 1.03032 to 1.02853, still above $42/41$ (`code/sector42_znam_violations.gp`, `code/sector42_znam_dp.rs`). The next step is therefore not a sieve but an exact enumeration, and it goes through.

Proposition 1.19 (The $d = 42$ sector at 64 primes)**.** *No solution of #307 has $a = 42q$ with q prime and $\omega(b) = 64$. Hence in that sector $\omega(b) \geq 66$ and $|P \cup Q| \geq 70$.*

Proof (finite verification). Write $e = b$, $T = 42/41$, and let $A_1 < A_2 < \cdots$ be the primes other than $2, 3, 7, 41$, $S_j = \sum_{i \le j} 1/A_i$. By the sector identity, $\sigma(e) = T - 1764/(41e)$, and $e > 3.5 \times 10^{57}$ by Theorem 4.3, so $T - 10^{-55} < \sigma(e) < T$. Exactly, $A_{64} = 337$, $A_{197} = 1229$, $A_{198} = 1231$, and $S_{61} + 3/1231 < T$ (`code/sector42_k64.gp`). If m of the 64 primes of e exceed 1229 then $\sigma(e) \le S_{64-m} + m/1231$, which for $m \ge 3$ is below $S_{61} + 3/1231 < T - 10^{-55}$; so $m \le 2$. Next, if $\operatorname{supp}(e) = S \cup \{\ell\}$ with $R = \prod S$, the identity $42e - 41e' = 1764$ reads $\ell\,\Delta_{42}(R) = 41R + 1764$, so $\Delta_{42}(R) > 0$ and ℓ is the integer $(41R + 1764)/\Delta_{42}(R)$: the last prime is determined by the others.

Case $m \le 1$. Take ℓ the prime exceeding 1229 if $m = 1$, else the largest prime of e, so $\ell \ge A_{64}$. The other 63 primes S lie among $A_1, \ldots, A_{197}$ with $\sigma(S) = \sigma(e) - 1/\ell \in [T - 1/A_{64} - 10^{-55}, T)$. The enumeration tests 36,901,596,763 sets, every such S among them; for none is $(41R + 1764)/\Delta_{42}(R)$ an integer.

Case $m = 2$. Let $\ell_2 \le \ell_1$ be the two primes exceeding 1229, S the other 62, and $\delta = T - \sigma(S)$. Then $1/\ell_1 + 1/\ell_2 \in (\delta - 10^{-55}, \delta)$, so $\delta < 2/1231$ and $\ell_2 \in (1/\delta, 2/\delta + 1)$. The enumeration tests 166,213 sets, every such S among them; the least δ among them is 1.539×10^{-6}, so every ℓ_2 range is finite (largest upper bound 1,299,387). For each S and each prime ℓ_2 in its range, $S \cup \{\ell_2\}$ determines ℓ_1 as above; it is never an integer.

The enumeration prunes on the mass in floating point with a margin of 10^{-9} on both ends of each window, so no set in a window is lost; the integrality test is a double–double prefilter with a conservative band followed by exact integer arithmetic (7,454,365 and 2,495 exact tests respectively), the integer layer checked against PARI to the digit (`code/sector42_k64.rs`, two runs with independent big–integer implementations). The non–enumerative half of the argument (the forced–prime identity and its converse, the sign condition, the case–split inequality and the parity step) is formalised in `lean/Erdos307/Sector42.lean`, with the two finite computations appearing as explicit hypotheses of the concluding statement. So $\omega(e) \ne 64$. By [6], $\omega(e) \ge 64$ and $\omega(e)$ is even (e is odd and $e' = 42q$ is even); hence $\omega(e) \ge 66$. Finally $q \notin \{2, 3, 7\}$ and $q \nmid e$, since $\gcd(a, b) = 1$ and $a = 42q$, so the four listed primes are distinct and $|P \cup Q| = |\{2, 3, 7, q\}| + \omega(e) \ge 70$. □

Proposition 1.20 (The primary pseudoperfect sectors)**.** *Let d be squarefree and* $\operatorname{supp}(e) = S \cup \{\ell\}$ *with $R = \prod S$. The sector equation $de - d'e' = d^2$ reads*

$$\ell\big(dR - d'R'\big) \;=\; d'R + d^2,$$

so the last prime is determined by the others in every sector, not only at $d = 42$. Running the resulting enumeration at the first admissible $\omega(e)$ above the mass floor gives, in each case, no solution:

d	$\omega(e)$	*sets*	$\lvert P \cup Q\rvert$ *before*	*after*
42	62	175	66	68
47058	54	41,654	60	62
2214502422	70	11,646	77	79

(`code/sector_kexclude.rs`; the identity is checked symbolically and in exact rationals in `code/sector_kexclude.g`
At each $\omega(e)$ in the table the mass bound $S_{\omega-1} + 1/A_{N+1} < d/d'$ excludes even one prime beyond the truncation, so there a single phase is a complete argument. At $d = 47058$ this raises a primary pseudoperfect sector off the global barrier: that sector sat at exactly 60*, and $|P \cup Q| \ge 62$ there. With Proposition 1.19 the $d = 42$ sector stands at $|P \cup Q| \ge 70$.*

The condition that makes a phase count sufficient is worth isolating, because it prices every rung of the ladder in advance and because a phase–1 enumeration that returns nothing proves nothing without it.

Proposition 1.21 (When the sector enumeration is complete). *Let d be squarefree, $d' = \sum_{p|d} d/p$, $t = d/d'$, and let $A_1 < A_2 < \cdots$ be the primes dividing neither d nor d', with $S_m = \sum_{i\le m} 1/A_i$. Fix a truncation N and a target $\omega(e) = k$. If j of the primes of e exceed A_N then*

$$\sigma(e) \ \le\ S_{k-j} + \frac{j}{A_{N+1}},$$

so whenever $S_{k-j}+j/A_{N+1} < t$ no admissible e has exactly j primes beyond the truncation. Writing $J = J(k,N)$ for the least such j, every admissible e has at most $J-1$ primes beyond A_N, and the phases $1,\dots,J-1$ exhaust the possibilities: the enumeration is complete.

Proof. The $k-j$ primes of e not exceeding A_N lie among $A_1,\dots,A_N$, so their reciprocals sum to at most that of the $k-j$ smallest, namely S_{k-j}; each of the remaining j is at least A_{N+1} and contributes at most $1/A_{N+1}$. A sector solution requires $\sigma(e) \ge t$, which the displayed bound then contradicts. Since the bound is decreasing in j once S_{k-j} falls below t, the values $j \ge J$ are all excluded together. $\square$

At the floor $k = K(d)$ one has $J = 1$, and there phase 1 alone is a complete argument, which is the case the table above uses; at $k = K(d)+2$ one has $J = 3$, and there phases 1 and 2 together are complete. Both correspondences are verified directly at $d = 47058$, at $k = 54$ and $k = 56$ respectively. Since only those two phases are implemented, the reachable rungs are those with $J \le 3$, and J is what prices the ladder:

d	$\omega(d)$	$K(d)$	k	J	$\lvert P\cup Q\rvert$ *if excluded*
42	3	62	62	1	68
42	3	62	64	3	70
42	3	62	66	5	72
47058	5	54	54	1	62
47058	5	54	56	3	64
47058	5	54	58	6	66
2214502422	6	70	70	1	79
2214502422	6	70	72	3	81

at $N = 400$ (`code/sector_phasecount.gp`). *The truncation is worth minimising rather than maximising: J is what must stay small, and at $d = 2214502422$ with $k = 72$ it equals 3 for every $N \ge 300$ and 4 below that, so $N = 300$ is the smallest complete choice and the one that leaves the fewest subsets to enumerate. The value of J grows quickly with k, and raising the truncation barely helps, since j/A_{N+1} falls only as fast as A_{N+1} grows: at $d = 42$ the rung $k = 66$ needs $J = 5$ at every N from 400 to 3000. That is what stops the $d = 42$ sector at $|P\cup Q| \ge 70$, and it is a limitation of the implemented phases rather than of the method.*

The next step at $d = 47058$ costs more, because the single–phase condition fails there. With $N = 400$ and $A_{N+1} = 2791$ one has $S_{55} + 1/A_{N+1} = 1.00629$ and $S_{54} + 2/A_{N+1} = 1.00309$, both above $d/d' = 1.0000213$, so one and two primes beyond the truncation must each be enumerated; but $S_{53} + 3/A_{N+1} = 0.99984 < d/d'$, so three or more cannot occur and the two phases together are complete. Both are empty: phase 1 enumerates 4.387×10^{11} leaves and 88,472,925 exact checks, phase 2 a further 373,644 leaves and 59,276 exact checks, and neither yields an admissible forced prime (`code/sector_kexclude.rs`, 12,658 s on nine cores). *Hence $\omega(e) \ne 56$. Since d is even, $e = d + qd'$ is odd and $e' = dq$ is even, so Proposition 1.23 forces $\omega(e)$ even; with $\omega(e) \ge 56$ already in hand this gives $\omega(e) \ge 58$ and*

$$|P\cup Q| \ =\ 1+\omega(d)+\omega(e) \ \ge\ 1+5+58 \ =\ 64$$

at $d = 47058$. The truncation is what stops the ladder here: at $\omega(e) = 57$ even four primes beyond A_N remain possible, so the next rung needs a larger N rather than more time at this one.

Remark 1.22. The $d = 42$ row recovers [6]'s $\omega(e) \geq 64$ by a route with nothing in common with its proof: that one maximises mass over a 51,281-state residue automaton, this one enumerates 175 prime sets and tests an integrality condition. Two independent derivations of the same bound.

Running the same exclusion across sectors rather than up the ladder is what the identity really buys. Among all sectors d that are squarefree with $\omega(d) \leq 12$ and every prime factor at most 71, exactly 16,234 have mass floor $|P \cup Q| = 60$; in every one of them the floor value of $\omega(e)$ is excluded, the largest enumeration being 41,654 sets (`code/sector_at60.gp`, `code/sector_kexclude.rs`). The counts 250, 1,533 and 16,234 are reproduced from scratch by `code/sector_at60_count.rs`. Determining them requires the complete factorisation of d', not merely its small factors: a prime of d' that goes unfound is never matched in the exclusion set, so the primes available to e are overcounted, $K(d)$ comes out too small and the floor too low, which is the direction in which a sector at 60 could be misread as lying below 60 and skipped. Since d' passes 10^{14} already at $\omega(d) = 10$ and routinely carries a large prime factor, the enumerator factors it with Pollard rho under a Miller–Rabin test, and refuses outright when d' would exceed the width its modular arithmetic is exact to. So

$$|P \cup Q| = 60 \text{ is impossible in every such sector,}$$

and any support of size 60 must have $\omega(d) \geq 13$ or a sector prime exceeding 71. This does not move Theorem 4.3, whose 60 is proved by a different route and over all supports; what it does is remove the entire low–ω, small–prime part of the sector decomposition from the level–60 frontier.

The sectors that could reach 60 form a *finite* set, which is worth recording because the sector decomposition appears to range over unboundedly many d. At level 60 the support is a 59–prime base together with one tail prime, and a base element x satisfies $T_{58} + 1/x \geq 2$, so $x \leq 1/(2 - T_{58}) = 793.6\ldots$; every prime up to 167 lies in the base, and d is a subset of the 138 primes at most 787. So the question "which sectors admit $|P \cup Q| = 60$" is a finite one. It is not, however, a small one: the count of sectors whose mass floor is exactly 60 rises from 250 over primes at most 47 with $\omega(d) \leq 8$ to 1,533 over primes at most 59 with $\omega(d) \leq 10$ and 16,234 over primes at most 71 with $\omega(d) \leq 12$ (all cleared, each family a subset of the next), and it does not saturate: raising the cap one step further, to primes at most 89 at the same $\omega(d) \leq 12$, gives 125,813, a factor of 7.75 for one more prime in the pool. Those are counted, not cleared, and are recorded here only as the next term of the growth. Holding $\omega(d) \leq 8$ and letting the prime bound B grow, the count runs 26, 83, 250, 588, 1762, 5172 at $B = 31, 41, 47, 59, 71, 89$; holding $\omega(d) \leq 6$ it runs 19, 34, 53, 77, 107, 166, 217 at $B = 31, 41, 47, 59, 71, 89, 107$. Against $\pi(B)$ the fitted exponent grows with ω, from about 2.7 at $\omega = 6$ to about 6.8 at $\omega = 8$, so the count behaves like a binomial $\binom{\pi(B)}{\omega}$. With the true bounds $B = 787$ and $\omega(d) \leq 58$ the family is finite but of order $\binom{138}{58}$, and enumerating it is not possible. Clearing sectors one at a time therefore strengthens the statement at each step and cannot reach the barrier. Finiteness here converts the sector programme into a computation of the same order as the one it was meant to replace.

Across every sector enumerated so far the floor value of $\omega(e)$ is not attained, and that is a falsifiable statement in its own right.

Proposition 1.23 (Odd sectors keep the prime 2 only when $\omega(d)$ is odd)**.** *Let d be odd and squarefree. Then $d' \equiv \omega(d) \pmod 2$. Consequently, if $\omega(d)$ is even then $2 \mid d'$, so $2 \mid dd'$ and $2 \notin \operatorname{supp}(e)$ by Remark 1.18; the mass d/d' must then be carried entirely by odd primes dividing neither d nor d'.*

Proof. For odd squarefree d each cofactor d/p is a product of odd primes, hence odd, so $d' = \sum_{p|d} d/p$ is a sum of $\omega(d)$ odd terms and has the parity of $\omega(d)$. Verified with no violation on the 81,053 odd squarefree $d < 2 \times 10^5$. □

The cost of losing 2 is not marginal, because $\frac{1}{2}$ is the largest single contribution available: at $d = 15015 = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, where $\omega(d) = 5$ is odd and 2 survives, $K(d) = 127$, while at $d = 255255 = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, where $\omega(d) = 6$ is even and 2 is lost, $K(d) = 1535$. Hence a sector with d odd and $\omega(d)$ even has mass floor far above 60, and every sector of mass floor 60 with d odd has $\omega(d)$ odd. That is what the enumeration shows: of the 16,234 sectors of mass floor 60 over primes at most 71, the 4,846 with d odd have $\omega(d) \in \{9, 11\}$ and none has $\omega(d)$ even, although nothing in the enumeration imposes it.

Conjecture 1.24 (The floor gap)**.** *For every squarefree d, no two–cycle in the sector d has $\omega(e)$ equal to the mass floor $K(d)$, the least k with $S_k(d) \geq d/d'$.*

The evidence is 5,680 sectors with no attained floor: the 1,533 of mass floor 60 over primes at most 59, a further 3,600 of the family over primes at most 71, the three sectors of Proposition 1.20, and 480 sectors of mass floor 61 through 68 over primes at most 47, the last of these a range disjoint from all the others and tested after the statement was fixed. Its consequence is that the barrier would move: the minimum of $1 + \omega(d) + K(d)$ over sectors is 60, attained at $d = 2 \cdot 3 \cdot 7 \cdot 23 \cdot 41$ and checked over 109,293 sectors, so with the conjecture and the parity of Proposition 1.20 every sector would give $|P \cup Q| \geq 62$. The evidence must be read with its power, which is negligible. The integrality $\Delta_d(R) \mid d'R + d^2$ has probability of order $1/\Delta_d(R)$, and $\Delta_d(R)$ is of the size of a product of some sixty primes; over the 5,680 sectors and their some 10^4 candidate sets each, the expected number of attained floors is of order 10^{-92} whether the conjecture is true or false. An enumeration of this kind therefore cannot support it, and no feasible enlargement would. Nor is the floor structurally distinguished: the rate at which candidates reach the exact divisibility test matches the width of the numerical prefilter at the floor and above it alike, the ratios of observed to expected being 0.72 and 0.86 at the floor for $d = 47058$ and $d = 2214502422$ and 1.01 at $\omega(e) = 64$ for $d = 42$, so the deficit at the floor is the small number of candidate sets and nothing else. The conjecture is recorded because it is what every enumeration performed here verifies and because its consequence is sharp, not because the enumerations are evidence for it.

The ladder does not continue cheaply. In every sector the first rung above the mass floor is small, the three counts above, and the next requires the two-phase machinery of Proposition 1.19, because the truncation bound no longer excludes primes beyond A_N: at $d = 42$ that step cost 3.7×10^{10} sets, and at $d = 47058$ the analogous $\omega(e) = 56$ is 4.4×10^{11}, not run. The cheapness of the first rung and the cost of the second is the shape of the method, not an accident of $d = 42$.

Remark 1.25. The same argument decides $\omega(e) = 66$ in principle, but with up to four primes beyond the truncation the nested ranges multiply, and the enumeration is no longer a twenty–minute job; we have not run it. What the proposition shows is where the sector programme actually stops: not at the residue sieve, whose limit is 68, but at the cost of an exact enumeration in which the last one or two primes are forced by the rest. The two–large–prime case, a factoring problem in general ($(\Delta\ell_1 - 41R)(\Delta\ell_2 - 41R) = 1681R^2 + 1764\Delta$), is finite here only because the smaller of the two is trapped in $(1/\delta, 2/\delta)$ and δ is bounded below over a finite list.

Three consequences, in increasing order of what they cost.

The gate is finite, not Lehmer. The tail primes of a base lie in the explicit finite set $\{2(N \pm k)/m^2 : m \mid 4D,\ k^2 = AB + Dm^2\}$. So the second of the three gates is a search over the divisors of $4D$, not a primality question along an exponential sequence, and the classification of the residual difficulty as Lehmer–class rather than Schinzel–class does not survive. Checked sound *and complete* against exhaustive search on 1200 bases with zero mismatches, and stepped out along a real orbit, in `code/tailbound.gp`: on the base $D = 1$, $N = 7$ (bound $N = 7$) the orbit gives $q_1 = 7$ prime, then $q_2 = 741895$, $q_3 = 76921173511$, and every later term composite, with both square conditions holding throughout.

The count reverses sign, and the search has now been run. The expected number of admissible m is $\sum_{m|4D}(AB + Dm^2)^{-1/2} \approx D^{-1/2} \sum_{m|4D} m^{-1}$, which on the first immune base is 2.4×10^{-56} and below 10^{-38} summed over all 34 immune families, the opposite of the "weakly favours YES" of Remark 1.13 on exactly the families that reading was about. That prediction is now checked. The full divisor space is $4 \cdot 2^{58} \approx 1.15 \times 10^{18}$ per family and is not enumerable, but because the hit probability falls like $1/(m\sqrt{D})$ the expected count is carried by small m: restricting to $\omega(m) \leq 8$ leaves 99.9998% of the expected count. Over all 34 families that slice is 3.0777×10^{11} candidates, and

$$\text{it contains no } m \text{ with } AB + Dm^2 \text{ a perfect square}$$

(4,933 survive the modular cascade, exact `issquare` kills all of them; the earlier $\omega(m) \leq 7$ pass, 4.7081×10^{10} candidates and 771 survivors, agrees).

The same search has been run over *every* admissible base, which matters because Proposition 4.7 requires no primality of A_S or B_S, only the algebra. The 34 immune families are the ones left after the Jacobi kill and the factoring attack, and their immunity is conditional on a Baillie–PSW verdict; the other 3,444 of the 3,478 open families are open precisely because A_S or B_S is a balanced semiprime nobody can split. The tail search does not care. Over all 49,961 admissible bases at $\omega(m) \leq 4$, which is 99.2% of the expected count per base, the run is 9.1296×10^{10} candidates, 1,469 survive the modular cascade, and exact `issquare` kills all 1,469. Every survivor was additionally checked to satisfy $m \mid 4D$, the hypothesis the proposition needs. So

$$\text{no admissible base admits a Pythagorean tail prime with } \omega(m) \leq 4,$$

unconditionally and with no primality input. Where Proposition 1.2 leaves 3,478 of the one–new–prime families at level 60 open, of which 34 only conditionally immune, this leaves none open on the stated slice. It is a slice result and not a proof: the untested remainder is 0.8% of the expected count per base, and $\omega(m) \geq 5$ is untested. The run is `code/tailsearch.rs` (a cascade of quadratic–residue filters modulo primes coprime to $2D$, 771 survivors) with every survivor then tested exactly by `issquare` in `code/tailsearch_verify.gp` (0 squares). A positive control recovers the known solutions of the toy base $D = 1$, $N = 7$, and an exhaustive exact run at $\omega(m) \leq 2$ agrees with the filter, so the cascade is not silently over–rejecting.

It is worth recording what the slice covers in the other measure, since the two are thirteen orders apart. The parametrisation runs over $m \mid 4D$ with m even, so $m = 2\alpha$ with $\alpha \mid D$, and at level 59 that is $2^{59} = 5.7646 \times 10^{17}$ divisors. Splitting on the parity of α, the divisors with $\omega(m) \leq 4$ number $2\sum_{j\leq 3}\binom{58}{j} = 65{,}136$, and those with $\omega(m) \leq 8$ number $2\sum_{j\leq 7}\binom{58}{j} = 692{,}376{,}800$. As fractions of the space the bound $q \leq N$ actually proves finite, these are

$$1.130 \times 10^{-13} \qquad \text{and} \qquad 1.201 \times 10^{-9},$$

against the 99.2% and 99.9998% of the *expected count* quoted above. Both figures are correct and they measure different things: the expected count concentrates on small ω, the hit probability falling like $1/(m\sqrt{D})$, so a slice can carry almost all of the expectation while carrying almost none of the space. Only the second pair of numbers is a statement about what has been checked. This is VERIFIED on the stated slice, not on the full divisor set: the residual 0.0018% of the expected count is untested.

The filter primes must be coprime to $2D$, which is itself a structural fact rather than a convenience. Since D contains every prime up to 167, for any $\ell \mid D$

$$AB + Dm^2 - N^2 = D\bigl(m^2 - 4D\bigr) \equiv 0 \pmod{\ell},$$

so the value is the square N^2 modulo ℓ for *every* m, and as $\gcd(D, N) = 1$ Hensel lifts that root to every power of ℓ. The tail search therefore has no local obstruction at any prime dividing D, the analogue of Proposition 1.14 for this formulation; the classical 64/63/65/11 pre–filter for squares rejects nothing here, measured at a 75% pass rate modulo 64 and 100% modulo 63. Formalised as `tail_value_mod_dvd` and `hensel_lift_identity`; the induction from the step to all powers is routine and is not formalised.

The paper's own height estimate now says the same thing. Remark 1.26 puts the least point of a nonempty fiber at height $\exp\bigl(10^{112}\bigr)$ on regulator grounds, so its $q \approx x^2/A$ carries some 10^{112} digits, against a bound of 115. Two independent estimates, one from divisor counting and one from the class–number formula, agree that no immune family admits a Pythagorean tail prime. Neither is a proof: the bound $q \le N$ is proved, the emptiness is HEURISTIC.

Formalised in `lean/Erdos307/TailBound.lean` (`tail_factor`, `tail_quadratic`, `tail_discriminant`, `tail_bound`, `tail_finite`) on the three standard axioms, with a non–vacuity witness at the base above.

Remark 1.26 (The immune signal and its Pell orbit)**.** The orbit below is real, but primality along it is not the residual question, since Proposition 4.7 bounds the tail prime by N.

For a reciprocity–immune base the Pell system of Remark 1.1, $B_S x^2 - A_S y^2 = -4D_S^2$, is locally soluble everywhere ($\kappa_S > 0$), and a Pythagorean pair in the tail family $S \cup \{q\}$ is *exactly* a solution with $q = (x^2 - D_S)/A_S = (y^2 - D_S)/B_S$ a positive prime. Its solutions grow like ε^n in the fundamental unit of the order of discriminant $4A_SB_S$ (224 digits, A_SB_S a nonsquare, on an explicit immune base). Writing $x_{n+1} = tx_n - x_{n-1}$ with $x_n = (\alpha^n + \beta^n)/2$ and $\alpha\beta = 1$, the squares x_n^2 have characteristic roots $\alpha^2, \beta^2, 1$ and the constant shift is absorbed by the root 1, so

$$q_n = \frac{x_n^2 - D_S}{A_S} \qquad \text{satisfies} \qquad q_{n+3} = (t^2 - 1)q_{n+2} - (t^2 - 1)q_{n+1} + q_n,$$

a non–degenerate ternary linear recurrence (verified on four Pell parameters). A reciprocity–killed base is the complementary degeneration: there the same system is locally *insoluble* ($\kappa_S = 0$), so no candidate exists at all, and the dichotomy of Proposition 1.2 is Pell–solubility versus Pell–insolubility.

The return is milder than "$k = 0$" suggests: on the Pythagorean locus the deviation $k = (a' - b)/a$ is an *integer* (the identity $v(D(u) - v) = u(u - D(v))$ with $\gcd(u, v) = 1$), forced into $\{0, -1\}$ by the near–balanced immune ratio, so $k = 0$ is a sign choice rather than a measure–zero coincidence, and the naive Wieferich shortcut that would pin q is vacuous, since $x^2 \equiv D_S \pmod{q}$ holds identically from $x^2 = A_S q + D_S$.

Neither square condition carries the difficulty alone. Imposing only the minus–square, $d^2 = B_S q + D_S$, the admissible d satisfy $d^2 \equiv D_S \pmod{B_S}$, and along such a class $d = d_0 + tB_S$ one has

$$q(t) \;=\; \frac{d_0^2 - D_S}{B_S} \;+\; 2d_0\, t \;+\; B_S\, t^2,$$

a quadratic in t (toy bases: $3 + 54t + 173t^2$, $7 + 238t + 1693t^2$, $2 + 446t + 17357t^2$): a Bunyakovsky question, of the same class as the $p = 5$ family of Proposition 1.15(iii), and conjecturally routine. It is the *conjunction* of the two squares that replaces the quadratic parameter by a fundamental unit and turns the orbit exponential.

Where the reading broke. That exponential growth was taken to place the question in the Lehmer class, on two grounds: no algebraic factorisation is available, D_S being squarefree so that $x^2 - D_S$ is irreducible, and no covering congruence can kill the family, by Proposition 1.14. Both observations are correct, and neither is the relevant one. The factorisation that does exist is not of $x^2 - D_S$ but of $x^2 - y^2 = 4D_S q$, available because $A_S - B_S = 4D_S$ *exactly*: the identity was used here only to build the Pell system, never multiplicatively. Proposition 4.7 draws it, and the ternary recurrence above then supplies no prime term past $q \le N$.

Three nested gates separate an immune family from a solution: fiber nonemptiness, a tail prime, and the $k = 0$ return. The first is a class–group condition, since Hasse supplies only rational points, and it is quantitative: immunity makes $A_S B_S$ a product of two distinct primes, hence squarefree, so the conic lives in the near–maximal order of $\mathbb{Q}(\sqrt{A_S B_S})$ of discriminant $\sim A_S B_S \approx 10^{224}$, and the class–number formula puts the expected regulator at $R = \sqrt{\Delta}\, L(1, \chi)/(2h) \approx 10^{112}$; barring an anomalously large class number, the *least* point of a nonempty fiber sits at height $\exp\big(10^{112}\big)$. That estimate is used in Remark 1.16 as independent corroboration that the second gate is empty. A direct sweep to $q \le 10^{12}$ was structurally incapable of reaching even the first gate.

Remark 1.27 (The affine sieve does not reach a rank–one orbit)**.** The modern instrument for primes in orbits is the affine sieve of Bourgain, Gamburd and Sarnak [46], which produces r–almost–primes in orbits of thin groups. It is the one method designed for exactly the shape of question Remark 1.26 isolates, so its failure here is worth recording precisely rather than left implicit.

The affine sieve needs a spectral gap: the congruence quotients of the acting group must form an expander family, and that is what supplies the level of distribution the sieve consumes. Here the q_n run along a Pell orbit, so the acting group is the cyclic group generated by the fundamental unit of the order of discriminant $4A_S B_S$. Cyclic groups are abelian, hence amenable, and have no spectral gap; expansion is precisely the property an abelian group lacks. The sieve therefore returns nothing, and it returns nothing on $2^n - 1$ for the same reason. The obstruction is rank, not thinness as such: non–elementary thin orbits, Apollonian packings among them, do yield almost–primes.

The natural repair fails for a reason worth stating too. One may embed the rank–one orbit in a larger thin group whose orbit does expand, and sieve there. What the affine sieve then delivers is infinitely many almost–primes in the *larger* orbit, inside which our sequence has relative density zero; the sieve does not localise to a prescribed sub–orbit. Expansion is bought at the cost of losing the sequence.

This meets the unit–rank trichotomy of Remark 1.14 from a second direction. There, rank zero gave a classical gate and denied a continuous family, positive rank the reverse. Here, rank one is exactly the cell in which the affine sieve has no traction. Stated as a single missing property: what is wanted is a substitute for expansion at rank one, and that is the Mersenne problem itself, not a lemma about #307.

Remark 1.28 (The gap to the record, measured)**.** It is worth stating how far the required statement is from the best that is known, because the distance is not one of degree.

Write a sequence's *density exponent* θ for $\#\{n \le X\} \asymp X^{\theta + o(1)}$. The sparsest sets currently proved to contain infinitely many primes are the values of binary forms: $a^2 + b^4$ at $\theta = 3/4$, by Friedlander and Iwaniec [41], and $a^3 + 2b^3$ at $\theta = 2/3$, by Heath-Brown [42], the latter being the sparsest polynomial form known to capture its primes. Our q_n grow like ε^{2n}, so $\#\{n : q_n \le X\} \asymp \log X$ and

$$\theta = 0.$$

The distance from $2/3$ to 0 is categorical rather than quantitative. Both records are bilinear arguments: they run a sieve of dimension one and close it with a Type II estimate obtained by averaging over a set of size a positive power of X. Every gain in such an argument is a gain of a power of X, and against $\log X$ terms a power of X gains nothing; below $\theta = 1/2$ the standard sieve axioms already fail, and at $\theta = 0$ there is no set left to average over. This is the same wall as Remark 1.27 approached from the analytic rather than the spectral side, and it is why every sequence of exponent zero, Mersenne, Fibonacci, Lucas and ours alike, is open together.

The structure our sequence does carry is the standard one and confers nothing. From $A_S q_n = x_n^2 - D_S$ one gets $x_n^2 \equiv D_S$ for every prime $p \mid q_n$ not dividing D_S, so all prime factors of q_n lie in the density-$\frac{1}{2}$ set where D_S is a quadratic residue. That constrains the factorisations without excluding any, which is exactly the position the primitive divisor theorems already left us in: divisors, never primality.

Remark 1.29 (The residue, atomised, and where its difficulty comes from)**.** Decomposing the question into independently known statements locates the obstruction on a single atom and identifies its source.

Two atoms carry theorems. Every q_n beyond the thirtieth has a primitive prime divisor [49]; and the greatest prime factor satisfies $P^+(q_n) > n \exp\bigl(\log n/(104 \log\log n)\bigr)$, by Stewart [43], which settled questions of Schinzel and Erdős and was the first general advance on Bang and Carmichael. Every other atom, that q_n be squarefree infinitely often, that $\omega(q_n)$ be bounded infinitely often, that q_n be a P_k, and primality itself, is open.

The chain that would close the crux is P^+ large, hence ω small, hence prime. It breaks at the first link, and by a measurable amount. Stewart's bound certifies $\log P^+ \geq (1 + o(1)) \log n$ against $\log q_n \approx 2n \log \varepsilon$, so the fraction of the term's size that is certified is of order $\log n/n$: 10.6% at $n = 10$, 0.32% at $n = 10^3$, 4×10^{-4}% at $n = 10^6$. Primality is the fraction 1. No sharpening of the constant 104 changes this, since $n^{1+o(1)}$ is the shape linear forms in logarithms produce. The Subspace Theorem is the other available instrument and does not help: it is stronger under weaker hypotheses but ineffective [44], and an ineffective statement cannot exhibit the single prime term a construction needs.

The source is visible in the comparison with the polynomial case. For $n^2 + 1$ the greatest prime factor exceeds $n^{1.3}$ infinitely often [45], that is $q^{0.65}$, a *positive proportion* of the term's size, and on such sequences almost-primality is available too. For an exponentially parametrised sequence only $(\log q)^{1+o(1)}$ is available, a vanishing proportion. The gap between the two regimes is exactly the gap between polynomial and exponential parametrisation, and Proposition 5.5 proves #307 admits no polynomial one. So the obstruction recorded in Remark 1.27 as a question of rank, in Remark 1.28 as one of density exponent, and here as one of P^+, is a single obstruction with three faces, and its source is Proposition 5.5: forced exponentiality is what places the residue in the class where nothing of the required strength is known.

Two things sharpen that attribution, and both make the obstruction less incidental rather than more. First, the boundary is exactly where Proposition 5.5 sits. The fraction of a term's size that Stewart's bound certifies is $\log P^+/\log q_n \approx \log n/\log q_n$, so it stays bounded below precisely when $\log q_n \ll \log n$, that is when q_n grows polynomially: degree d gives the constant $1/d$. Every super-polynomial rate sends it to 0, and not only the exponential one: $\exp(\sqrt{n})$ and $n^{\log n}$ do as well. So polynomial growth is not one convenient case among several, it is the whole of the favourable region, and it is exactly what Proposition 5.5 excludes.

Second, the hypotheses of Proposition 5.5 are worth testing, since a gap there would dissolve everything above. It assumes a *fixed* number of polynomials, and its proof uses that finiteness when it splits $\sum_i 1/p_i$ into constants plus a remainder tending to 0. A family whose support grows with the parameter is therefore not covered. That gap is real and useless: a growing support means a growing number of simultaneous primality conditions, which is a harder demand than one, not an easier one, and it does not return the family to the polynomial regime. The proposition forbids the case that would help and leaves uncovered only cases that would not.

Remark 1.30 (The squarefree atom, and what it reduces to)**.** Among the four open atoms one is structurally easier, and it is worth saying why and how far it gets. Squarefreeness of q_n asks only for an *upper* bound on a count of bad events, never for the production of a prime, so it is the one atom not subject to the parity obstruction that blocks bounded ω, P_k, and primality.

The argument it wants is a union bound. For a prime $p \nmid 2A_SD_S$, $p^2 \mid q_n$ forces $x_n^2 \equiv D_S \pmod{p^2}$; the Pell orbit is purely periodic modulo p^2 with period $\pi(p^2)$, there are at most two square roots of D_S, and each is met a bounded number of times per period, so the density of bad n attached to p is at most $C/\pi(p^2)$. Where $\pi(p^2) = p\,\pi(p)$, which is the generic behaviour, and $\pi(p) \asymp p$, this is $\asymp C/p^2$, and the union bound closes as soon as $C\sum_p p^{-2} < 1$. The constant is comfortable: $\sum_p p^{-2} = 0.4522474\ldots$, so any $C < 2.21$ suffices, and a positive proportion of the q_n would be squarefree, hence infinitely many.

What is missing is the hypothesis $\pi(p^2) = p\,\pi(p)$. Its failure is precisely a Wall–Sun–Sun prime for the sequence, and there the density is $\asymp 1/p$ rather than $\asymp 1/p^2$, whose sum diverges and destroys the bound. No such prime is known for any classical sequence and their finiteness is unproved, so the atom does not close: it *reduces* to a Wall–Sun–Sun statement. That is a sharper position than "open", since it names the obstruction and shows it is a single hypothesis rather than a programme, but it is a reduction and not a proof.

The unconditional pieces are formalised. `Erdos307.primeSquareSum_certificate` gives $\sum_p p^{-2} < \frac{1}{2}$ in the finite form used above, the first six primes together with the bound $\frac{1}{2}\sum_{m\geq 16} m^{-2} < \frac{1}{30}$ for the odd primes beyond; `good_density_pos` and `squarefree_positive_density_of_bound` are the union bound and the conditional implication. All carry 0 `sorry` and the three standard axioms (`lean/Erdos307/Squarefree.lean`).

Two further points fix the standing of this atom, and both are negative. First, on the immune bases the argument is blocked at its *lowest* rung as well as its highest. The density attached to a single small p is $\#\{n \bmod \pi(p^2) : p^2 \mid q_n\}/\pi(p^2)$, read off the Pell orbit modulo p^2, and that needs the fundamental solution, the object Remark 2.4 (i) places at $10^{24.7}$ operations. For a Lucas sequence with a known fundamental unit the small-prime rung is a finite computation; here it is not. The ladder is therefore blocked at both ends, by non-computability below and by Wall–Sun–Sun above. Second, and more to the point, squarefreeness is not what the construction needs: Remark 1.26 requires q to be *prime*, and a squarefree q with several prime factors is useless there, so even the dream lemma of this atom would leave #307 exactly where it stands. Of the four atoms this is the easiest, and it is also the only one whose resolution would not advance the problem.

2 Local anatomy and a heuristic model

Lemma 2.1 (Local anatomy)**.** *Let a be squarefree and q a prime with $q \nmid a$. Then $q \mid a'$ if and only if $\sum_{p\mid a} p^{-1} \equiv 0 \pmod{q}$, where p^{-1} is the inverse of p modulo q. If $q \mid a$, then $q \nmid a'$.*

Formalised. *`Erdos307.anatomy_iff`, with the factorisation $a' \equiv a\sum_{p\mid a} p^{-1}$ proved as `csum_eq_mul_recipSum` rather than assumed; the second half is `anatomy_on_support` and is Rigidity* (`lean/Erdos307/Frame.lean`).

Proof. $a' = a\sum_{p\mid a} 1/p$. If $q \nmid a$ then $q \mid a'$ iff $q \mid a\sum_{p\mid a} p^{-1}$ iff $\sum_{p\mid a} p^{-1} \equiv 0 \pmod{q}$. If $q \mid a$, then by Lemma 2.1 applied to the prime set of a, $q \nmid a'$. □

Thus the cycle equation $a' = b$, $b' = a$ factors, prime by prime, into $\omega(a)+\omega(b)$ local congruences of exactly the shape that governs amicable–pair heuristics: for each $q \mid b$ one needs $\sum_{p\mid a} p^{-1} \equiv 0 \pmod{q}$, and symmetrically. These local conditions, however, are individually *empty of obstruction*:

Proposition 2.2 (No single–modulus congruence obstruction). *For every modulus m with $2 \le m \le 210$, the reduced cross–match system $N_P \equiv D_Q$, $N_Q \equiv D_P \pmod{m}$ is realisable by disjoint squarefree prime sets P, Q, in both structural branches:* $\gcd(m, D_P D_Q) = 1$*, and the branch in which a prime divisor of m lies in $P \cup Q$. Hence no individual modulus in this range can rule out a solution of #307.*

Construction. Write $\Sigma_S \equiv \sum_{p \in S} p^{-1} \pmod{m}$, so $N_S \equiv D_S \Sigma_S$. The reduced system is $\Sigma_P \equiv D_Q D_P^{-1}$, $\Sigma_Q \equiv D_P D_Q^{-1} \pmod{m}$; its only coupling is $\Sigma_P \Sigma_Q \equiv 1$, automatic, with D_P, D_Q otherwise free. By Dirichlet each unit class mod m contains primes, fixing D_P, D_Q; appending a prime $r \equiv 1 \pmod{m}$ leaves D_S fixed while sending $N_S \mapsto rN_S + D_S \equiv N_S + D_S$, i.e. incrementing Σ_S by 1, so (when $\gcd(D_S, m) = 1$) every target class is reached with disjoint sets. In the branch $\ell \mid m$, $\ell \in P$, reduction mod ℓ collapses the system to $N_Q \equiv D_P \equiv 0 \pmod{\ell}$, met by inverse–paired residues in Q; composite m needs no separate gluing, the increment acting directly mod m. We verified both branches for all $2 \le m \le 210$ (for prime and prime–power moduli the inside branch requires the directed construction above; seeding a prime divisor of m into P, a blind search missing these as a search–power artefact, not an arithmetic obstruction). $\square$

Remark 2.3 (Precise scope). This excludes only a *single fixed–modulus* congruence disproof. The witnesses are m–dependent (the modulus–6 witness already fails modulo 210), and a single (P, Q) solving the system for *all* m simultaneously is, for bounded integers, equivalent to the integer identities $N_P = D_Q$, $N_Q = D_P$ (that is, to #307 itself) so per–modulus solvability is *no* evidence for existence. Nor does it exclude other finite or local negative arguments (size/Mertens bounds, density, magnitude–congruence hybrids, or a Hasse–type failure across infinitely many moduli); indeed our own $|U| \ge 59$ and 2.09×10^{56} barriers are finite–data results fully consistent with it. Together with Theorem 1.1 (every non–archimedean shadow trivially negative for degree reasons) it brackets the problem: no single–modulus obstruction over $\mathbb{Z}$, fatal non–archimedean analogues over $\mathbb{F}_q[t]$, the difficulty living at the archimedean place.

A first–order count. Model a candidate squarefree a as "P–abundant" ($\sum_{p \mid a} 1/p > 1$) and ask for $b = a'$ to itself be Q–abundant with $\sum_{q \mid b} 1/q = 1/s$. Two empirical inputs (Section 6) calibrate this:

- the density of P–abundant n is $\delta = 0.0420$ (and ≈ 0.0176 among squarefree n); a theorem, not merely an observation (Proposition 2.6);
- for a qualifying squarefree a, $b = a'$ is squarefree about 95% of the time, but its mass $\sum_{q \mid b} 1/q$ has mean ≈ 0.047 against the required ≈ 0.934; the empirical tail probability $\Pr[\sum_{q \mid b} 1/q \ge 1/s]$ is 0 in 40,920 samples at these scales (Section 6 of the core note).

The gap is structural: at small scale a P–abundant a hogs the cheap mass–carrying small primes, and $b = a'$, being coprime to a, is both too small and too coprime to assemble reciprocal mass $1/s$. The extremal case is stark: for the primorial $a = \prod_{i \le k} p_i$, though $\sum_{p \mid a} 1/p$ rises through 2 at $k = 59$, the cofactor sum a' is essentially prime, with $\sum_{q \mid a'} 1/q < 0.2$ for every $k < 100$ (and only 0.0014 at $k = 59$): the family that *maximises* the forward mass is the most starved on the return. Equivalently, for squarefree n with n' squarefree, $n'' = n\,\sigma(n)\,\sigma(n')$, so the two–step return ratio $n''/n = \sigma(n)\sigma(n')$ is the #307 product itself; over *all* such $n < 10^6$ it stays strictly below 1, never exceeding 0.54 (maximal at $n = 969969 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$), so D is two–step contracting throughout the tested range and must reverse to unit ratio for a solution to exist. The first scale at which b could plausibly carry the required mass is of the order of the barrier ($\sim 10^{56}$); this dynamical mass–gap is *consistent with* the proved barrier rather than an independent derivation of it (the empirical tail probability 0 at small scale is equally consistent with NO and with YES–beyond–the–barrier). A naive "expected number of solutions" built from the local congruences (one factor $\sim 1/q$ per condition) diverges like a constant times $\log X$; this is a divergent naive count of exactly the type known to be unreliable, so we treat it only as very weak, non–rigorous suggestive evidence that solutions could exist at or beyond the barrier.

Remark 2.4 (Calibration)**.** We present the count of the previous paragraph as a heuristic, not a theorem. Heuristics of amicable–pair type are known to be delicate; we report it as weak evidence, consistent with the divergence of the refined expected count, that the answer to #307 is plausibly YES (and hence that the period–one form of the Ufnarovski–Åhlander conjecture is plausibly false).

Remark 2.5 (A cautionary refinement)**.** One can try to sharpen this into a quantitative rate using a Kubilius/Erdős–Wintner model, stratified over which small primes lie in the support of a versus remain available to b. Two lessons emerge, both methodological. First, a plain Monte–Carlo estimate *undercounts* the rate by some 10^{61}: the expectation is dominated by rare, balanced splits of the small primes that sampling almost never sees, so a naive estimate of this kind is misleadingly small and only an explicit stratified sum corrects it. Second, and this is why we record no numerical prediction, the stratified estimate is itself unstable: refining the stratification from the first 8 to the first 14 primes drops the rate by some 17 orders of magnitude and pushes the implied first occurrence from $\sim 10^{10^{38}}$ out past $\sim 10^{10^{55}}$, with no sign of convergence. The only robust conclusion is negative: every version places a hypothetical first solution *doubly exponentially* beyond any conceivable search, consistent with the constructive deficit of Section 3; the model does not reliably decide existence in either direction.

Proposition 2.6 (The abundance density is a theorem)**.** *The additive function $f(n) = \sum_{p|n} 1/p$ satisfies the Erdős–Wintner three–series criterion [19] (here $\sum 1/p^2$ and $\sum 1/p^3$ converge), so f has a limiting distribution and the density $\delta = \lim_{x\to\infty} \frac{1}{x}\#\{n \le x : f(n) > 1\}$ exists, equal to $\Pr\left[\sum_p X_p/p > 1\right]$ for independent $X_p \sim \mathrm{Bernoulli}(1/p)$. Moreover the constant carries a* certified *enclosure:*

$$0.04202 \ \le\ \delta\ \le\ 0.04223.$$

Writing $S_1 = \sum_{p\le P_1} X_p/p$, $P_1 = 3\times 10^4$, and $T \ge 0$ for the tail, monotone upper and lower cdf envelopes of S_1 are propagated through the Bernoulli convolution on a grid of step 10^{-7} with the shift $1/p$ rounded down for the upper and up for the lower envelope (which preserves them, cdfs being nondecreasing) so $\delta \ge 1 - U(1)$ costs nothing (the tail is nonnegative), while $\delta \le 1 - L(1 - a) + \Pr[T \ge a]$ with the explicit Chernoff bound $\Pr[T \ge a] \le \exp(1.36 - P_1 a)$, $a = 10^{-3}$ (from $\log \mathbb{E}\, e^{\lambda T} \le \lambda s_2 + 0.72\lambda^2 s_3$ at $\lambda = P_1$, using $s_2 < 1/P_1$, $s_3 < 1/(2P_1^2)$ and $e^x - 1 \le x + 0.72x^2$ on $[0, 1]$); float slack 10^{-9} exceeds the accumulated rounding by nine orders, and the machinery reproduces an exact four–prime reference to 2×10^{-16} (`code/delta_cert.py`). The point value 0.0420*, at the bottom of the enclosure, agrees to four places with the sieve to 2×10^7; the upper edge is conservative, carried almost entirely by the a–window. As for the density of abundant integers, no closed form is expected.*

Remark 2.7 (Two dynamical probes: no second obstruction)**.** Two measurements sharpen the picture beyond the first–order mass count, and both cut towards rather than against a solution. *(i) Image density.* A two–cycle needs *both* members in the image $D(\mathbb{N})$ (each is the other's derivative), so one might fear a second obstruction; that the high–mass numbers a cycle requires are seldom derivatives. The opposite holds: of the integers below 10^7, 68.3% lie in $D(\mathbb{N})$, and the fraction *rises* with mass, from 61.7% at $\sum_{p|n} 1/p < 0.2$ to 78.2% at $\sum_{p|n} 1/p > 1$ and 80% beyond. The high–mass numbers are thus over–represented among derivatives; the second cycle condition is comfortably met. *(ii) The barrier is where the balanced mass–product reaches* 1. Set $r(n) = \left(\sum_{p|n} 1/p\right)\left(\sum_{q|n'} 1/q\right) = D^2(n)/n$, so $r = 1$ is exactly the cycle condition. Exhaustively over $n \le X$ the maximum of r climbs $0.50 \to 0.62$ as X runs $10^3 \to 10^9$ (with no exact cycle, as the barrier demands), with no ceiling; the climb follows the balanced split $r \approx (S(k)/2)^2$, $S(k) = \sum_{i \le k} 1/p_i$, which reaches 1 precisely when $S(k) = 2$, i.e. $k \approx 59$. The dynamical trajectory thus reproduces the $|U| \ge 59$ barrier from the opposite direction. Neither probe decides existence; both *remove* hypothetical obstructions, leaving the exact archimedean closure as the sole gate.

Remark 2.8 (Why the analytic method is starved: $N(r) \le 1$)**.** The rank–one rigidity (Section 9 of the core note) has a sharp consequence for the circle–method machinery that resolves the *unrestricted* Egyptian–fraction problem. For a rational r let $N(r) = \#\{\, P : \sum_{p\in P} 1/p = r \,\}$ count the finite prime sets with reciprocal sum r. Then

$$N(r) \le 1 :$$

by the Rigidity Lemma (Theorem 2.1) the sum $\sum_{p\in P} 1/p = N_P/D_P$ is already in lowest terms, so the reduced denominator of r equals $D_P = \prod P$ and P is exactly its set of prime divisors; uniquely determined ($N(r) = 1$ precisely when that denominator is squarefree and the numerator of r is its cofactor sum). The circle method needs the opposite: a target hit by a positive proportion of subsets, so a main term dominates. This is exactly what Bloom's shift to $\{p-1\}$ manufactures, $p-1$ being composite and divisor–rich, a rational is the reciprocal sum of *many* subsets of $\{1/(p-1)\}$ (already among the first fourteen shifted primes the multiplicity reaches 3 and grows), whence [7] represents every positive rational there. Over the primes themselves $N(r) \le 1$ leaves nothing to average: the analytic engine that reaches $\{p - h\}$ is provably starved on $\{p\}$, and this is the method–level reason #307 (the rank–one, exact–prime case) lies beyond it.

3 A breeder form and the feasibility of certificate search

Allowing *two* free primes on one side turns the divisor trick into a cleaner bilinear form, Euler's amicable–pair method, in the shape used by te Riele's breeding algorithms [14], and lets us state precisely the strongest honest conclusion: a *conditional* reduction, which nonetheless does not become a feasible finite search.

Proposition 3.1 (Bilinear breeder form)**.** *Fix coprime squarefree M_0, N and seek a two–cycle $a = M_0 r p$, $b = Nq$ with r, p, q distinct primes coprime to M_0N. Put $G := NM_0 - N'M_0'$ $(= \Delta_0)$, $c := M_0N'$, and $K := GN^2 + c^2$. Then the cycle equations $a' = b$, $b' = a$ are equivalent to*

$$(Gr - c)(Gp - c) = K, \qquad q = \frac{M_0 rp - N}{N'}.$$

Proof. Expanding, $(Gr - c)(Gp - c) - K = -N'G\,(a' - b)$ (verified symbolically), so the bilinear equation is equivalent to $a' = b$; the second cycle equation $b' = a$ then determines q as stated. □

Proposition 3.2 (The breeder is integral)**.** *Keep the notation of Proposition* 3.1 *and let $G \ne 0$. Then*

$$(Gr - c)(Gp - c) = K \iff G\,rp - c(r + p) = N^2,$$

and every solution of that equation has

$$N \mid a', \qquad q = \frac{M_0 rp - N}{N'} = \frac{a'}{N} \in \mathbb{Z}.$$

Proof. Expanding, $(Gr-c)(Gp-c)-c^2 = G\big(G\,rp - c(r+p)\big)$ while $K - c^2 = GN^2$, so the two equations differ by the factor $G \neq 0$. Substituting $G = NM_0 - N'M_0'$, $c = M_0N'$, $a = M_0rp$ and $a' = M_0'rp + M_0(r+p)$ into $G\,rp - c(r+p) = N^2$ gives $NM_0rp - N'M_0'rp - M_0N'(r+p) = N^2$, that is

$$N\,(a-N) \;=\; N'\big(M_0'rp + M_0(r+p)\big) \;=\; N'a'.$$

Hence $N \mid N'a'$, and N squarefree gives $\gcd(N, N') = 1$, so $N \mid a'$ and $q = a'/N$. $\square$

This is worth isolating because the displayed $q = (M_0rp - N)/N'$ is a quotient by a quantity of the size of the barrier, and one might expect a candidate to be usable only when N' happens to divide $M_0rp - N$, a further demand of probability about $1/N'$, which would leave the construction hopeless for reasons having nothing to do with primality. There is no such demand. The expectation $\mathbb{E} \approx \big(\tau(K)/G\big)(\ln r \ln p \ln q)^{-1}$ is therefore the correct count: the breeder loses candidates only to primality, never to integrality. A scan of 51,132 frames (`code/breeder_integral.gp`) produced 9 admissible candidates, every one with integral q. Formalised in `lean/Erdos307/Breeder.lean`.

Each factorisation $K = d \cdot (K/d)$ with $d \equiv -c \pmod{G}$ proposes a candidate $(r,p) = \big((d+c)/G, (K/d+c)/G\big)$; a solution is any such candidate whose r, p, q are all prime. A dual, partition–side form is occasionally useful:

Proposition 3.3 (Subset form)**.** *For a support U with $D = \prod_{p\in U} p$ and $\Sigma = \sum_{p\in U} D/p$, a partition $U = P \sqcup Q$ solves #307 iff $S = P$ satisfies $A(\Sigma - A) = D^2$, where $A := \sum_{p\in S} D/p$. Moreover $A(\Sigma - A) - D^2 = -(1 - s_Ps_Q)\,D^2$.*

(Both identities verified symbolically and on 2×10^4 random supports.) The last identity says the "analytic" residual $1 - s_Ps_Q$ and the "arithmetic" residual $A(\Sigma - A) - D^2$ are one equation up to the factor D^2: the two obvious steering handles carry no independent information.

A conditional reduction. Several ingredients of a construction are unconditional. Greedy Egyptian–fraction choice gives disjoint P, Q with $0 < 1 - s_Ps_Q = O(1/T)$ (solvability to any precision); a prime r nearest M_0N'/Δ_0 gives $G \le \Delta_0^{0.475}(M_0N')^{0.525}$ unconditionally (Baker–Harman–Pintz [15]), and each small prime ℓ can be forced to divide K by one congruence on a free prime slot of N (Dirichlet then supplies the prime). Granting these, #307 reduces to a single prime–density statement,

(H) among the admissible candidates so produced, some triple (r,p,q) is prime,

and (H) $\Rightarrow$ #307 is YES. This (H) is a Hardy–Littlewood/Bateman–Horn–type assertion for an explicit, computable, *non–polynomial* family; we know of no named conjecture implying it. That is not an accident of our construction:

Proposition 3.4 (No fixed–shape family, of any growth rate)**.** *Let $P(t) = \{p_1(t), \dots, p_k(t)\}$ and $Q(t) = \{q_1(t), \dots, q_l(t)\}$ be families of integers of* any *shape (no polynomial, algebraic or analytic hypothesis is imposed) indexed by $t \in \mathbb{N}$, with k, l fixed. Suppose each member takes positive prime values, each nonconstant member tends to $+\infty$, and $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for* infinitely many *t. Then every member is constant.*

Proof. Let φ enumerate, in increasing order, the t at which the relation holds; φ is strictly monotone, so $\varphi(n) \to \infty$. Split each side into its constant members, contributing A_P and A_Q, and its nonconstant ones, contributing $\varepsilon_P(t), \varepsilon_Q(t) > 0$ which tend to 0 because each nonconstant member tends to $+\infty$ and there are finitely many. Along φ the relation holds at every index and the remainders still tend to 0, so Proposition 5.5's two steps apply verbatim: the limit gives $A_PA_Q = 1$, and subtracting leaves $A_P\varepsilon_Q + A_Q\varepsilon_P + \varepsilon_P\varepsilon_Q = 0$ with every term nonnegative and some term strictly positive if a nonconstant member exists. □

The proof of Proposition 5.5 uses polynomiality in exactly one place, to pass from "holding at infinitely many t" to "holding identically", and restricting to the subsequence removes that need. What the argument actually consumes is three hypotheses (finitely many members, positive values, nonconstant members growing) and none of them is about shape.

The strengthening is not cosmetic. Excluding polynomial families rules out a reduction of #307 to Bunyakovsky, Schinzel or Bateman–Horn, but it says nothing about *exponential* families, and those are what the classical technique for problems of this kind actually uses: Thābit's rule for amicable pairs runs on primes $3\cdot 2^n - 1$, $3\cdot 2^{n-1} - 1$, $9\cdot 2^{2n-1} - 1$, and Euler's generalisation is of the same shape. A member $3\cdot 2^n - 1$ is positive and tends to infinity, so it satisfies the hypotheses above and the family is excluded. Proposition 3.4 therefore closes the whole Thābit/Euler programme for #307, not merely its polynomial shadow: no fixed–shape rule of any growth rate can generate solutions, and the constructive calculus of Section 5 of the core note is not failing for want of a cleverer ansatz.

Two limits on the statement should be read with it. The hypothesis that the families have *fixed* cardinality is used, in the splitting into constants plus a remainder tending to 0, and a family whose support grows with t is not covered; that gap is real, and harmless, since a growing support is a growing number of simultaneous primality demands rather than a weaker one. And positivity is a genuine hypothesis, not a consequence, for the reason recorded in Remark 3.5. Formalised as `Erdos307.no_family_of_any_shape` on three standard axioms, in the same shape as Proposition 5.5: the abstract core is proved, and the two facts the family supplies, that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to 0, enter as hypotheses rather than being derived, so what is imported stays visible. The theorem's statement quantifies over arbitrary index types and arbitrary $f_P : \mathbb{N} \to \iota \to \mathbb{R}$; no polynomial occurs anywhere in it.

Remark 3.5 (Positivity is a hypothesis, not a consequence)**.** It is not enough to ask only that each polynomial take prime values for infinitely many integers t, without requiring the values to be positive on a common branch. As printed that statement is *false*: take $P = (2, 2, t, -t)$ and $Q = (2, 2)$. Each member takes prime values for infinitely many integers, and

$$\left(\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{t} - \tfrac{1}{t}\right)\left(\tfrac{1}{2} + \tfrac{1}{2}\right) = 1$$

identically, with two nonconstant members present. The proof's step "every term on the left is nonnegative" silently assumes positivity, which is exactly the missing hypothesis; with it, as now stated, the proof is correct and the counterexample is excluded.

The formalisation carries the repaired statement, not the printed one: `Erdos307.nonconstant_empty` takes $0 < f_P(i)$ for every nonconstant member as an explicit hypothesis, so the Lean development was already stating the corrected proposition. This is the one place in the paper where formalisation caught a missing hypothesis before review did.

This upgrades the earlier computational observation (that the breeder family is not polynomially parametrised) to a theorem for *all* fixed–shape families: the prime–density hypothesis behind #307 is irreducibly non–polynomial, which is why no standard conjecture applies.

Proposition 3.6 (Near–miss frames are extremal for G)**.** *Let a be squarefree with a' squarefree, and take the frame $(M_0, N) = (a, a')$, so that $M_0' = a'$ and $N' = a''$. Then*

$$G \;=\; NM_0 - N'M_0' \;=\; a'\,(a - a''),$$

and since $a'' \neq a$ unless #307 is solved, $|a - a''| \geq 1$ and therefore $|G| \geq a'$.

Proof. Direct substitution; the inequality is $|a-a''| \geq 1$ for the nonzero integer $a-a''$. $\square$

The point is not that the bound is large but that it runs the wrong way. Writing $r = a''/a$ for the near–miss ratio, $G = aa'(1-r)$, so a frame built from a good near–miss has $1-r$ small and aa' correspondingly large, and the two cancel *exactly*, because $a(1-r) = a - a''$ is an integer and a nonzero one. Improving the near–miss therefore does not shrink G; it drives a' up and G with it. The witness of Proposition 2.9, with $|1-r| = 1.65 \times 10^{-13}$, gives $|G| \geq a' > 10^{112}$, against the $\log_{10} G \approx 13$ that the Baker–Harman–Pintz route reaches at the barrier. So the natural reading of "steer Δ_0 small by steering $s_{M_0} s_N$ to 1" is not merely circular in the sense of Section 5 of the core note: on near–miss frames it is inverted.

Small $|G|$ does occur, but only on small frames: over 1,143,888 coprime squarefree frames with $M_0, N \leq 2000$ the minimum is $|G| = 5$, at $(M_0, N) = (2, 3)$. That is the supply–demand conflict in raw form, and it can be measured. Binning by $|G|$ and taking the best frame in each bin (`code/breeder_supply.gp`) gives

$\log_{10}\lvert G\rvert$	0.5	1.5	2.5	3.5	4.5	5.0
$\max \tau(K)/\lvert G\rvert$	0.600	0.313	0.214	0.053	0.0065	0.0030

The ratio is largest on the smallest frames, where it is already below 1, and decays monotonically from there. Since a productive frame needs $\tau(K)/|G|$ well above 1, no frame in the range supplies one, and the trend is against scale rather than with it.

The descent, and why it cannot be iterated. Adjoining a prime w to M_0 acts on G *linearly*: with $M_0 \mapsto M_0 w$ one has $M_0' \mapsto M_0' w + M_0$ and hence

$$G \longmapsto wG - M_0 N', \qquad \text{so} \qquad |G_{\text{new}}| = |G| \cdot |w - w^*|, \qquad w^* := \frac{M_0 N'}{G} = \frac{\sigma(N)}{\varepsilon},$$

writing $\varepsilon = 1 - \sigma(M_0)\sigma(N) = G/(M_0 N)$. Since $G = 0$ is #307, this is a descent towards a solution, and on the ε scale it is quadratic: $\varepsilon_{\text{new}} = \varepsilon^2 \theta / \sigma(N)$ with $\theta = w - w^*$. It is a Newton iteration for #307.

Three facts close it. First, it cannot start below the barrier: $w^* = \sigma(N)/\varepsilon \geq 2$ needs $\varepsilon \leq \sigma(N)/2$, i.e. $\sigma(M_0)\sigma(N)$ near 1, which by AM–GM needs $\sigma(M_0 N) \geq 2$ and so $\omega(M_0 N) \geq 59$. Over 4,284,896 coprime squarefree frames with $M_0 \leq 5000$, $N \leq 3000$ the smallest $|\varepsilon|$ is 0.470, and $w^* < 2$ throughout.

Second, only $w = \lfloor w^* \rfloor$ or $\lceil w^* \rceil$ can shrink $|G|$, since $|G_{\text{new}}| < |G|$ demands $|w - w^*| < 1$. So a step descends only when one of those two integers is prime, with probability $\approx 2/\log w^*$.

Third, and decisively, a successful step squares the difficulty. From $w^* = M_0 N'/G$,

$$w^*_{\text{new}} = \frac{M_0 w N'}{G\theta} = w^* \cdot \frac{w}{\theta} \;\approx\; \frac{(w^*)^2}{\theta},$$

so $\log w^*$ more than doubles at every step while $\log |G|$ falls by only $|\log \theta| \approx 0.69$. The gain is linear in the number of steps k and the cost is quadratic in the exponent: $|G|$ falls like 2^{-k}, whereas the probability of surviving k steps is $\prod_{j<k} 2/\log w_j^* \asymp 2^{-k^2/2}$. Starting from the greedy frame $N = 30$, $M_0 = \prod_{7 \leq p \leq 271} p$, where $\log |G| = 249$ and $w^* = 428.2$ (`code/breeder_descent.gp`):

paths available	10^2	10^4	10^6	10^8	10^{10}	10^{12}
reachable depth k	3	5	6	7	8	8
$\lvert G\rvert$ shrinks by	8	32	64	128	256	256

against the factor 10^{108} required to reach $G = 0$. Even 10^{12} independent descent paths buy eight steps and a factor 256. This is the quantitative form of the circularity of Section 5 of the core note: the Newton iteration for #307 exists, is exact, and converges quadratically, and each step costs a prime whose size is the square of the last one's.

Why (H) **is not a finite search.** It is tempting to read (H) as "run a certificate hunt that is expected to succeed." It is not, for a quantitative reason. The expected number of certificates from one frame is $\mathbb{E} \approx \big(\tau(K)/G\big) \cdot (\ln r \, \ln p \, \ln q)^{-1}$, and *both* factors are pinned by the geometry.

- *Demand.* $G = \Delta_0 = M_0 N(1 - s_{M_0} s_N)$ exactly, so G is small only in the circular regime $s_{M_0} s_N \approx 1$; over 2×10^4 random coprime squarefree frames the ratio $G/(M_0 N)$ has median 1 and best value $\approx 10^{-0.23}$, and the honest Baker–Harman–Pintz reduction still leaves $\log_{10} G \approx 13$ at the barrier.

- *Supply.* $\tau(K)$ is bounded by the *size* of $K = GN^2 + c^2$ $(\geq c^2)$, which is an *output* of the frame, not a free dial. The maximal order of the divisor function [18, Thm. 317] caps $\log_2 \tau(K)$ at ≈ 36 for $K \leq 10^{54}$ (the barrier geometry) and at ≈ 112 even for $K \leq 10^{240}$; reaching $\tau(K) = 2^{120}$ needs $K \gtrsim 10^{263}$, i.e. frame scale far beyond the barrier.

Enlarging a frame grows $K \propto N^2$ but gains divisors only at rate $\log 2/\log\log K < 1$ per log–unit of K, while it grows G at rate 1 per log–unit of N; so $\tau(K)/G$ cannot be driven up. Optimising $\mathbb{E}$ over all (M_0, N) splits with geometry–consistent K and the maximal–order τ, and granting the most favourable demand and singular–series constants, yields $\log_{10} \mathbb{E} \leq -4$ at the barrier and *decreasing* with scale; empirically 385 factorable frames realise a median of 0 and a maximum of 1 admissible divisor, against the $\sim 10^5$ a single productive frame would need. A self–consistent accounting; capping the number w of forced prime factors of K by the frame's own residue freedom (counted generously), sharpens this to $\log_{10} \mathbb{E} \leq -8$ at $B = 56$, falling to ≈ -260 at $B = 3000$: the deficit is *scale–invariant*, so no choice of scale, slot count, or budget split reaches expectation 1. (The construction is even self–similar: engineering K's algebraic; e.g. Gaussian–integer; factorisation to supply divisors reduces to prescribing N and N' simultaneously, an arithmetic–antiderivative inverse problem of the same difficulty class as #307.) The breeder thus makes #307 conditional on (H) but does *not* render it a feasible finite computation: the supply–demand conflict through $K = GN^2 + c^2$ is exactly the circularity of Section 5 of the core note in quantitative form. There is also a structural reason the analogy with amicable pairs was bound to fall short: te Riele's breeding amplifies a *known seed* pair into new ones, whereas the barrier (Theorem 4.3) forbids any seed below 2.09×10^{56} the method imports a precondition the problem cannot supply.

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